ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems

- Discrete-Time Systems (Ref pp. 55-61)
- Classification of Discrete-Time Systems (Ref pp. 61-69)
- Linear Time Invariant (LTI) Systems (Ref pp. 71-100)
- Introduction to z-Transform (Ref pp. 151-174)
- Pole-Zero Description of Discrete-Time Systems (Ref pp. 174-183)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (Ref pp. 247-297)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

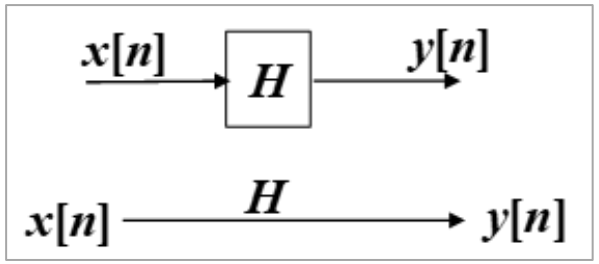
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Term 3, 2026

Discrete-time systems

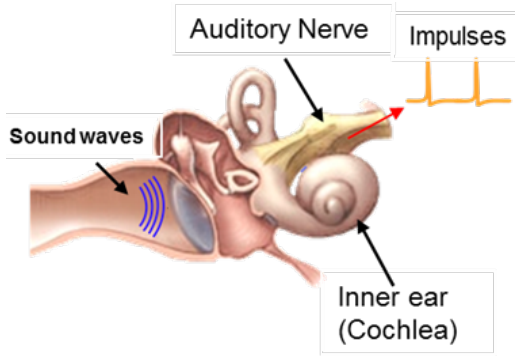
Introduction

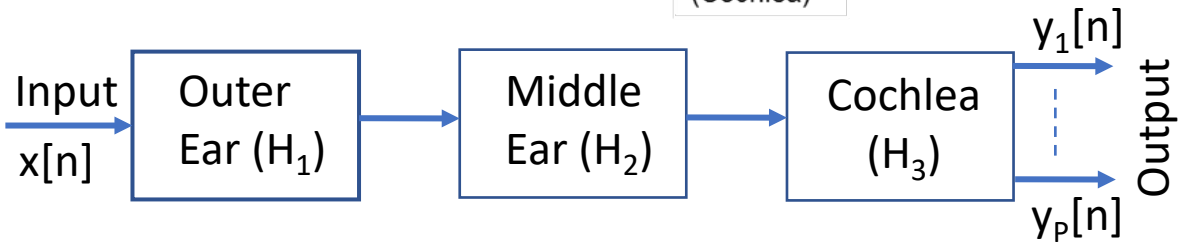
- A discrete-time system, *H*, is a device (or an algorithm) that operates on a discrete-time signal called the input $x[n]$ (or excitation) according to some well-defined rule, to produce another discrete-time signal, $y[n]$, called the output (or response) – see diagram below



- The transformation of the input signal $x[n]$ by the system H to produce the output signal $y[n]$ is expressed as: $y[n] = H\{x[n]\}$

Peripheral auditory system





For example, it is possible to implement a discrete-time model of the peripheral auditory system comprising the outer ear (H1), middle ear (H2) and cochlea (H3).

Unlike the simple system on the left with one input and one output, the cochlear model can produce multiple outputs corresponding to different frequency channels

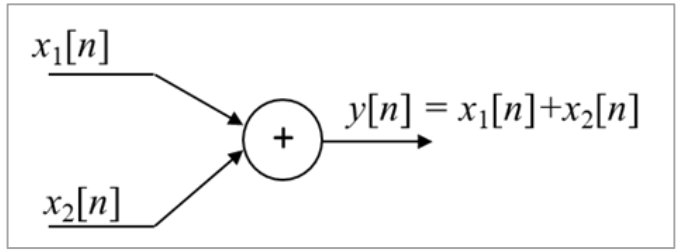
Discrete-time systems

Block Diagram Representation

- To introduce the block diagram representation of discrete-time systems, we first define some basic building blocks that can be interconnected to form more complex systems.

An adder

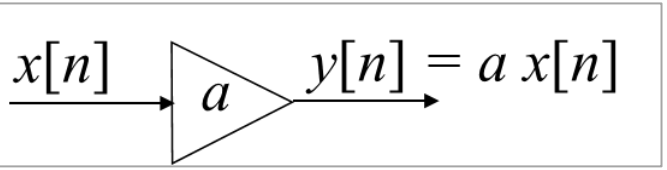
- An adder performs the addition of two signal sequences, $x_1[n]$ and $x_2[n]$ to produce the output sequence $y[n]$.



- Note that it is not necessary to store either of the input sequences to perform the addition. In other words, the addition operation is **memoryless**.

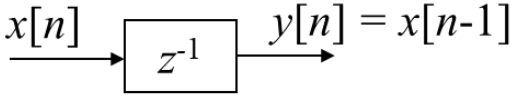
A constant multiplier

- This operation simply represents applying a scale factor 'a' to the input *x[n]*. Note that this operation is also **memoryless**.



A unit delay element

- The unit delay is a special system that simply delays the signal passing through it by one sample (or sample period).
- If the input signal is $x[n]$, the output is $x[n-1]$. The previous input sample $x[n-1]$ is stored in memory at time $n-1$ and it is recalled from memory at time n to form $y[n] = x[n-1]$.
- Thus, this basic building block **requires memory**. We use the symbol T or z^{-1} to denote a delay unit.



n	-1	0	1	2	3
x[n]	x[-1]=0	x[0]=1	x[1]=0	x[2]=5	x[3]=7
y[n]	y[-1]=?	y[0]=0	y[1]=1	y[2]=0	y[3]=5

Discrete-time systems

Difference Equations

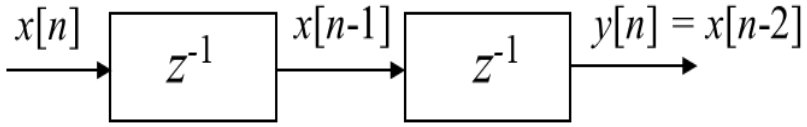
- A discrete-time system consisting of combinations of adders, multipliers and unit delays can always be described by a **difference equations**.
- The equations would be an ordinary algebraic equations if no delays were present.
- For example, the system described by:

$$y[n] = a_0 x[n] + b_1 y[n-1]$$

The present output $y[n]$ is the weighted sum of the previous output $y[n-1]$ and the present input $x[n]$. The quantities a_0 and b_1 are called *coefficients*.

Example 1:

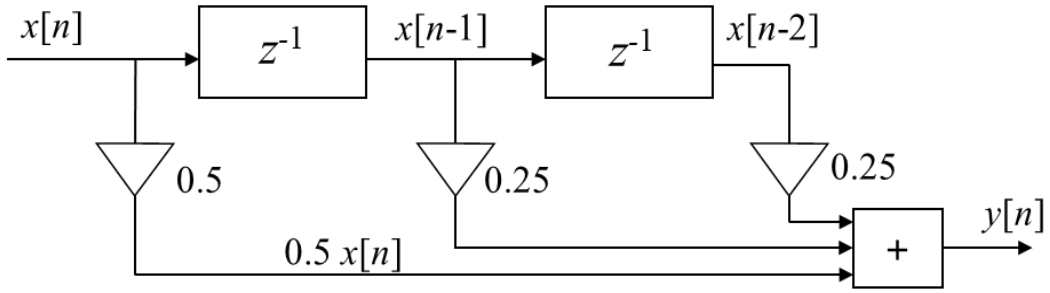
- Draw a system implementation for the difference equation: $y[n] = x[n-2]$



Example 2:

- Draw a system implementation for the difference equation:

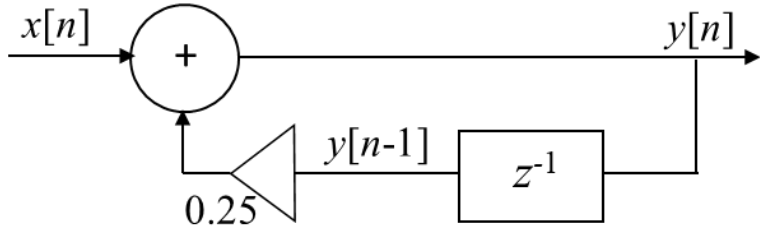
$$y[n] = 0.5\,x[n] + 0.25\,x[n-1] + 0.25\,x[n-2]$$



Example 3:

- Draw a system implementation for the difference equation:

$$y[n] = x[n] + 0.25y[n-1]$$



Discrete-time systems

Example 4:

- Draw a system implementation of the following the difference equation:

$$y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$$

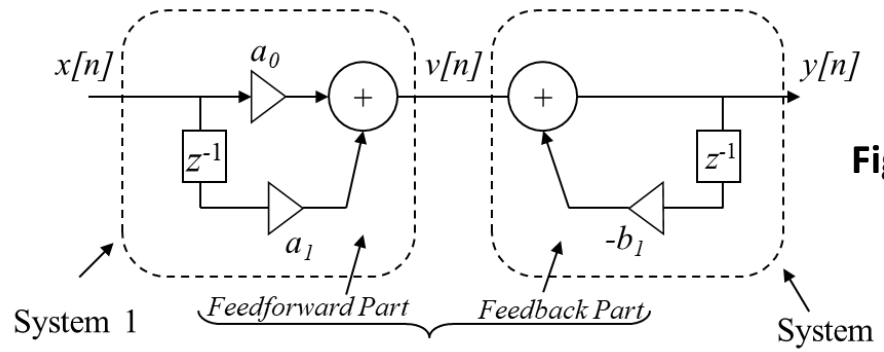


Figure 1

- Without Without changing the input-output relationship, the order of the two systems in the cascade can be interchanged.

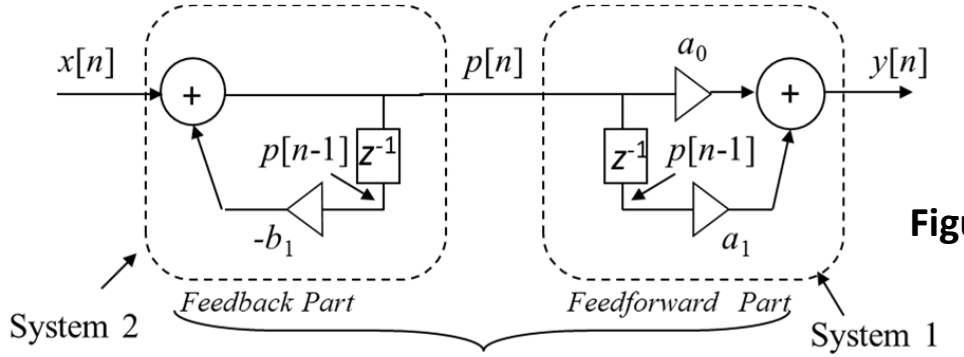
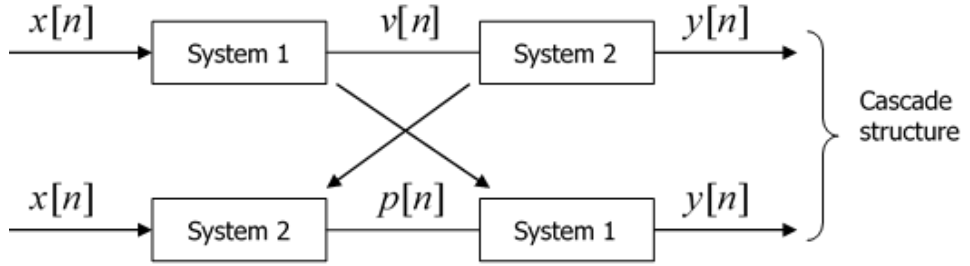


Figure 2

Note: For **LTI** systems, the order of systems connected in cascade can be interchanged without affecting the overall input-output relationship.



- The two delay elements can now be combined into a single delay, as shown below:

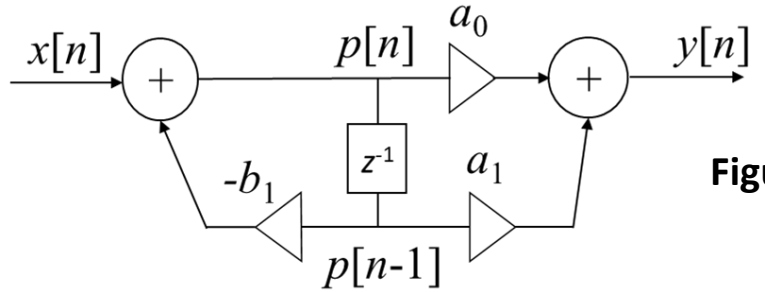


Figure 3

- It can be shown that Figures 1, 2 & 3 all represent the same difference equation:

$$y[n] = a_0 x[n] + a_1 x[n-1] - b_1 y[n-1]$$

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Classification of discrete-time systems

Classification of Systems

- In the analysis as well as in the design of systems, it is desirable to classify the systems according to the general properties they satisfy. General Categories are:
  1. Static systems
  2. Time - invariant systems
  3. Linear systems
  4. Causal systems
  5. Stable systems

1. Static systems

- A discrete-time system is called static or memoryless if its output at any instant '*n*' depends only on the input sample at the same time, but not on past or future input samples.

  For example, $y[n] = ax[n]$; $y[n] = nx[n] + bx^3[n]$ are both static or memoryless
- On the other hand, the systems described by the following input-output relations are dynamic systems or systems with memory:

  $y[n] = x[n] + 3x[n-1]$; $y[n] = \sum_{k=0}^{N} x[n-k]$

2. Time-invariant systems

- A time-invariant system is defined as follows:

  $x[n-n_0] \xrightarrow{H} y[n-n_0]$ where $y[n] = H\{x[n]\}$.
- A system is time-invariant if a time shift in the input produces the same time shift in the output.

3. Linear systems

- An input $x_1[n]$ to a system (H) results in an output $y_1[n]$; Given a different input $x_2[n]$, the system output is $y_2[n]$. $x_1[n] \xrightarrow{H} y_1[n]$; $x_2[n] \xrightarrow{H} y_2[n]$

For a system to be **linear**, when its input is $x_1[n] + x_2[n]$, the output must be the sum of the individual outputs

$$x_1[n] + x_2[n] \xrightarrow{H} y_1[n] + y_2[n]$$ **{Additivity}**

$$a_1 x_1[n] \xrightarrow{H} a_1 y_1[n]$$ **{Scaling}**

- A linear system is defined as follows:

  $$a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} a_1 y_1[n] + a_2 y_2[n]$$

  where a_1 and a_2 are arbitrary constants.

A linear system must satisfy both additivity and scaling.

Classification of discrete-time systems

4. Causal systems

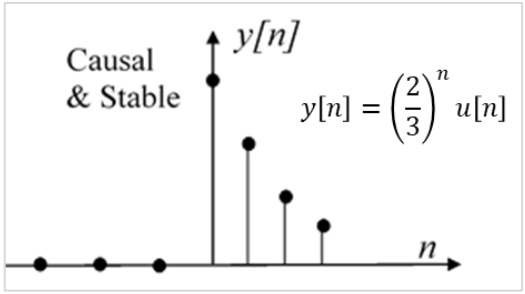
- A system is said to be causal if the output of the system at any time '*n*' depends only on present and past inputs, but does not depend on future inputs.
- If a system does not satisfy this definition, it is called *noncausal*. Such a system has an output that depends not only on present and past inputs but also on future inputs.

Examples:

$$y[n] = x[n] - x[n-1] \quad \rightarrow \text{Causal}$$
$$y[n] = ax[n] \quad \rightarrow \text{Causal}$$
$$y[n] = x[n] + 3x[n+4] \quad \rightarrow \text{Noncausal}$$
$$y[n] = x[-n] \quad \rightarrow \text{Noncausal}$$

- A discrete - time sequence is called (right-sided) **causal** if it has zero values for $n < 0$.

Note: For $y[n] = x[-n]$ *at* $n = -1$, $y[-1] = x[1]$, which depends on a future input. (Noncausal)



5. Stable systems

- A discrete signal $x[n]$ is bounded if there exists a finite *A* such that $|x[n]| < A$ for all *n*.
- A discrete-time system is Bounded Input-Bounded Output (**BIBO**) stable if every bounded input sequence $x[n]$ produced a bounded output sequence $y[n]$.

If $x[n]_{max} \leq A \; for \; all \; n$, then $y[n]_{max} \leq B \; for \; all \; n$

Example: The following discrete-time system is at rest {i.e. $y[-1] = 0$}. Check if the system is BIBO stable.

$$y[n] = ny[n-1] + x[n], \quad n > 0$$

Model answer:

If $x[n] = u[n]$, then $|x[n]| \leq 1$. But for this bounded input, the output is unbounded (see below).

$$n = 0 \rightarrow y[0] = x[0] = 1$$
$$n = 1 \rightarrow y[1] = 1y[0] + x[1] = 2$$
$$n = 2 \rightarrow y[2] = 2y[1] + x[2] = 5$$
$$n = 3 \rightarrow y[3] = 3y[2] + x[3] = 16$$
$$\vdots \qquad \vdots$$

Thus, the bounded input $x[n] = u[n]$ produces an unbounded output. Hence, the system is not BIBO stable.

Linear Time-Invariant (LTI) Systems

- A discrete-time system is called a Linear Time-Invariant (LTI) system if it satisfies both:
  1. **Linearity** – the system satisfies additivity and scaling.
  2. **Time invariance** – a time shift in the input produces the same time shift in the output.

- Thus, if $x[n] \xrightarrow{H} y[n]$ then for an LTI system:

- **Linearity:** $a_1 x_1[n] + a_2 x_2[n] \xrightarrow{H} a_1 y_1[n] + a_2 y_2[n]$

- **Time invariance:** $x[n - n_0] \xrightarrow{H} y[n - n_0]$

- LTI systems are particularly important in DSP because they can be completely characterised by their impulse response $h[n]$.

- LTI systems form an important class of systems in digital signal processing and will be studied in more detail in the following sections.

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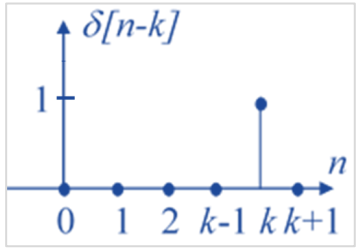
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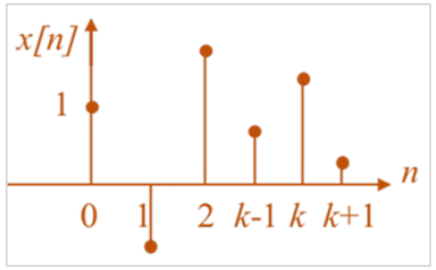
Transformation of Discrete Time Signals

- A discrete-time signal, $x[n]$ may be shifted in time (delayed or advanced) by replacing the time index n with n-k or $n + k$ where $k > 0$ is an integer .

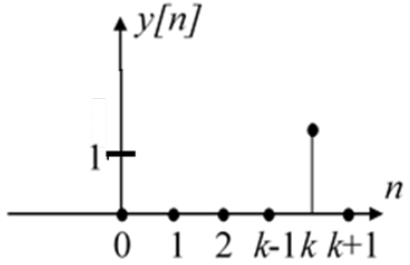
  $x[n-k]$ => $x[n]$ is delayed by k samples
  $x[n+k]$ => $x[n]$ is advanced by k samples

- For example, consider a shifted version of the unit impulse function and an arbitrary signal x[n] both shown below .



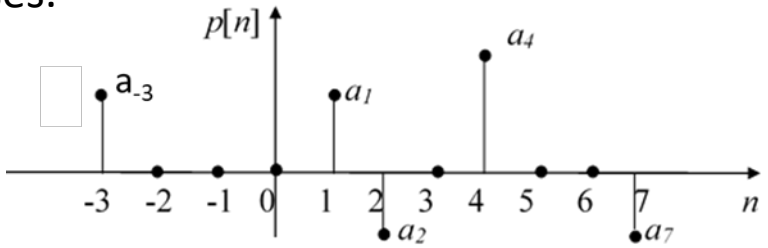


- If we multiply x[n] by $\delta[n-k]$, we obtain a signal that is zero everywhere, except at n = k.



∴ y[n] = x[n] · δ[n-k]
= x[k] · δ[n-k]

- An arbitrary sequence can then be expressed as a sum of scaled and delayed unit impulses.



$$p[n] = a_{-3}\delta[n+3] + a_1\delta[n-1] + a_2\delta[n-2] + a_4\delta[n-4] + a_7\delta[n-7]$$

- More generally, the discrete-time sequence can be expressed as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

- For a finite-length discrete-time signal with N samples::

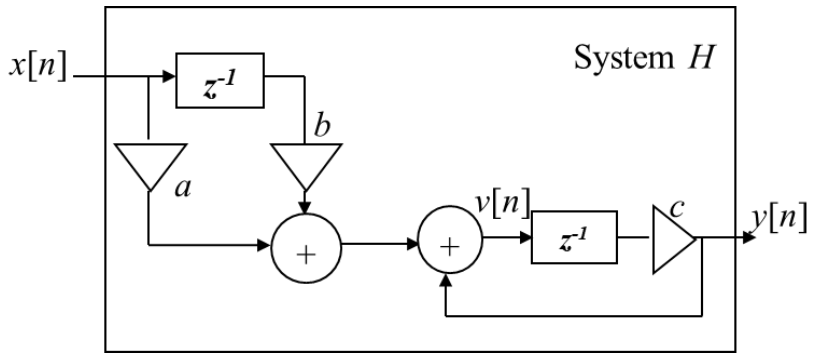
$$x[n] = \sum_{k=0}^{N-1} x[k]\,\delta[n-k]$$

- Generally, the **convolution** of two sequences $x[n]$ and $h[n]$ is defined as:

$$x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

Impulse Response of an LTI System

- For example, consider the discrete-time system, H, shown below.



- The difference equation for the system *H* is given by:

$$v[n] = ax[n] + bx[n-1] + y[n]$$

$$y[n] = cv[n-1]$$

- Assume that the system is initially at rest, i.e., $v[n] = 0$ and $y[n] = 0$ for n < 0 $\Rightarrow y[0] = 0$
- From the above difference equations, $y[n]$ can be determined for a given input.

$$\text{Let } x[n] = \delta[n] \quad \text{(Unit impulse)}$$

Substituting n = 0, 1 ,2 ,... in the above equation , we obtain,

$\boldsymbol{n = 0}$ $v[0] = a\,x[0] + b\,x[-1] + y[0] = a\cdot 1 + b\cdot 0 + 0 = a$

$\Rightarrow \boldsymbol{y[1]} = c \cdot v[0] = \boldsymbol{a.c}$

$\boldsymbol{n = 1}$

$v[1] = a\,x[1] + b\,x[0] + y[1] = a\cdot 0 + b\cdot 1 + a\cdot c = b + a\,c$

$\Rightarrow \boldsymbol{y[2]} = c\,v[1] = \boldsymbol{c\,(b+ac)}$

$\boldsymbol{n = 2}$

$v[2] = a\,x[2] + b\,x[1] + y[2] = 0 + 0 + c\,(b+ac)$

$\Rightarrow \boldsymbol{y[3]} = c\,v[2] = \boldsymbol{c^2\,(b+ac)}$

.....

$\boldsymbol{n = n - 1}$

$v[n\text{-}1] = c^{\,n-2}(b+ac) = bc^{n-2} + ac^{n-1}$

$\therefore \boldsymbol{y[n]} = c\,v[n\text{-}1] = \boldsymbol{bc^{n-1} + ac^{n}}$

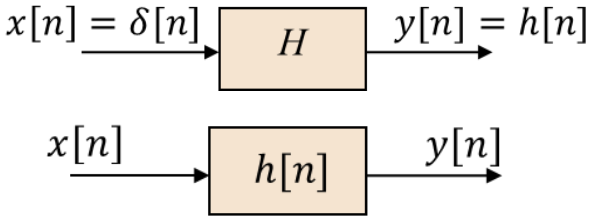
$$\therefore y[n]\Big|_{x[n]=\delta[n]} = \boldsymbol{h[n]} = bc^{n-1} + ac^{n} \quad n \geq 1$$

$$\boldsymbol{h[n]} = (b + ac)c^{n-1}\,u[n-1]$$

- The response of an LTI system to a unit impulse $x[n] = \delta[n]$ is called the impulse response and is denoted by $h[n]$. **An LTI system is completely characterised by its impulse response.**

LTI Systems: Impulse Response and Convolution

- If $x[n] = \delta[n], then\ y[n] = h[n]$. The output $h[n]$ is called the impulse response and completely characterises the LTI system H.



- The response to an arbitrary input signal $x[n]$ is the **convolution** of $x[n]$ with the impulse response *h[n]* of the system as given by the equation below:

$$y[n] = x[n] * h[n]$$

Convolution and commutative property

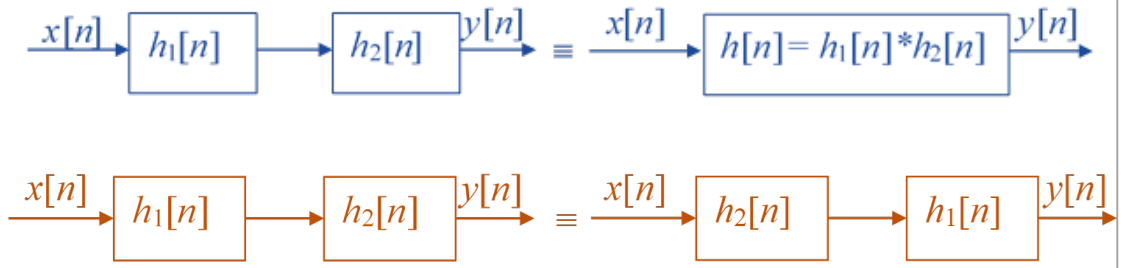
$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

where *h[n]=H{δ[n]};* $x[n] * h[n] = h[n] * x[n]$

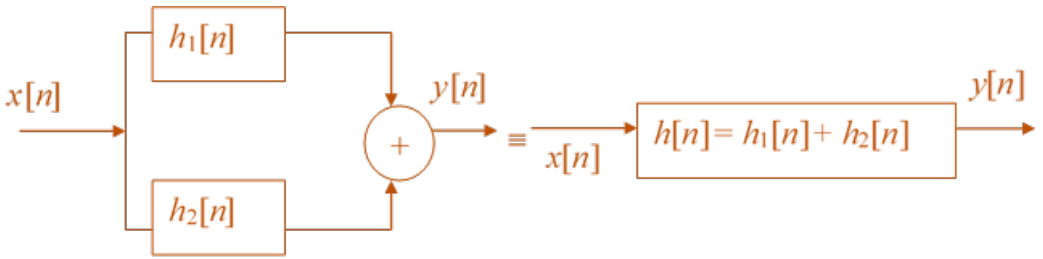
Associative Law

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$



Distributive Law

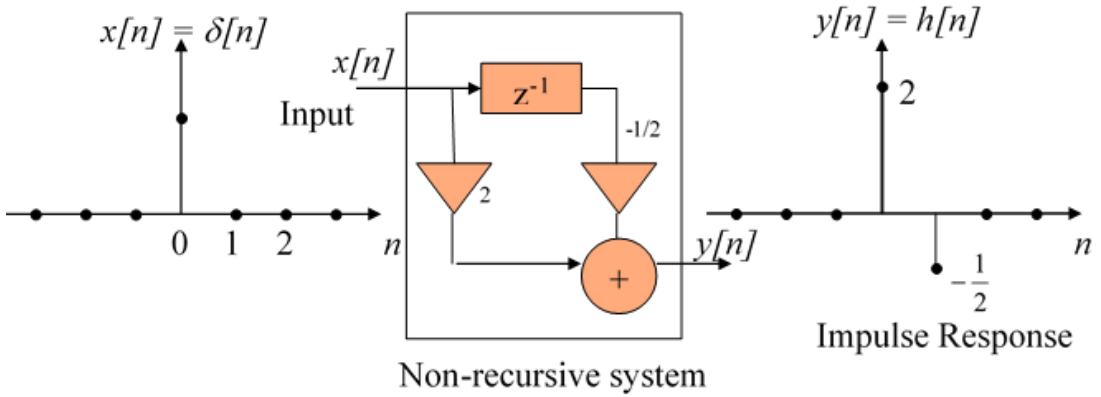
$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$



Impulse Response of a LTI System

Finite Impulse Response (FIR) System

- If the impulse response of a LTI system is of finite duration, the system is said to be Finite Impulse Response (FIR). It is also called **FIR filter.**



- Consider the system above; The difference equation is given by $y[n] = 2x[n] - 0.5\,x[n-1]$. Find the impulse response, $h[n]$, of the system. *Let* $x[n] = \delta[n]$

$$\boldsymbol{n=0} \quad y[0] = h[0] = 2\,x[0] - 0.5\,x[-1] = 2$$
$$\boldsymbol{n=1} \quad y[1] = h[1] = 2\,x[1] - 0.5\,x[0] = -0.5$$
$$\boldsymbol{n=2} \quad y[2] = h[2] = 2x[2] - 0.5x[1] = 0$$

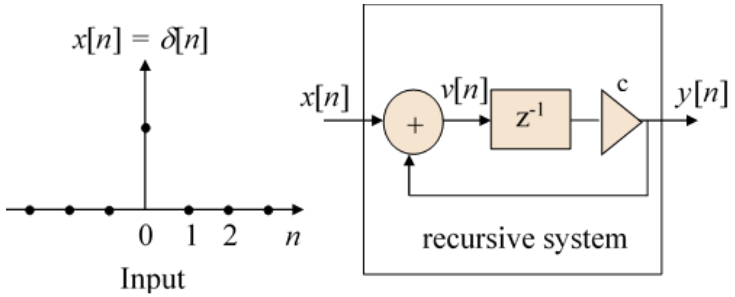
.....

$n = n$ y[n] = 0

$$h[n] = \{2, -0.5\}$$
$$h[n] = 0 \quad n \geq 2$$

Infinite Impulse Response (IIR) System

- If the impulse response of a linear time-invariant system is of infinite duration, the system is said to be an Infinite Impulse Response (IIR) system. It is also called **IIR filter.**



- Consider the system above; The difference equation is given by $v[n] = x[n] + y[n]$; $y[n] = c.v[n-1]$ $(0 < c < 1)$ Find the impulse response, $h[n]$, of the system. *Let* $x[n] = \delta[n]$

$\boldsymbol{n=0}$ $v[0] = x[0]+y[0] = 1$; $\therefore$ $\boldsymbol{y[1]} = c \cdot v[0] = \boldsymbol{c}$

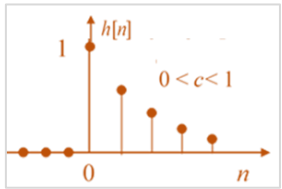
$\boldsymbol{n=1}$ $v[1] = x[1] + y[1] = c$; $\therefore \boldsymbol{y[2]} = c\,v[1] = \boldsymbol{c^2}$

$\boldsymbol{n=2}$ $v[2] = x[2] + y[2] = c^2$; $\therefore \boldsymbol{y[3]} = c\,v[2] = \boldsymbol{c^3}$

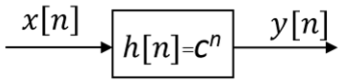
.....

$\boldsymbol{n = n-1}$ $v[n-1] = c^{n-1}$;

$\therefore$ $\boldsymbol{y[n]} = c\,v[n-1] = c.\boldsymbol{c^{n-1}}$



$\therefore \boldsymbol{h[n]} = c^n$ for $n \geq 0$ $\therefore \boldsymbol{h[n]} = c^n\,u[n]$



Examples

Example 1

- Find the impulse response $h[n]$ of the following fourth order non-recursive system.

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + a_3x[n-3] + a_4x[n-4]$$

Answer: To find $h[n]$, we let $x[n] = \delta[n]$.

n=0 ➔ $h[0] = a_0\delta[0] + a_1\delta[-1] + a_2\delta[-2] + a_3\delta[-3] + a_4\delta[-4] = a_0$

n=1 ➔ $h[1] = a_0\delta[1] + a_1\delta[0] + a_2\delta[-1] + a_3\delta[-2] + a_4\delta[-3] = a_1$

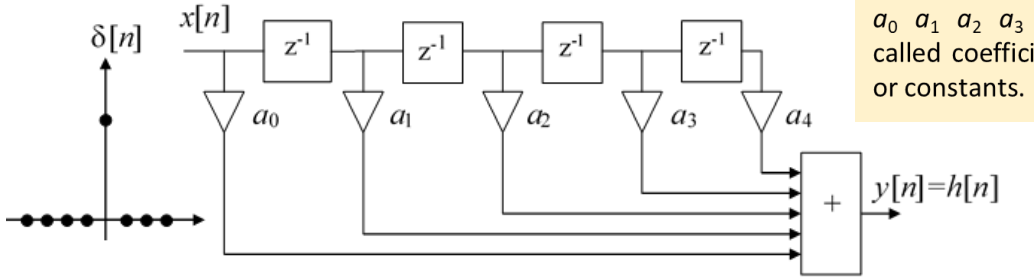
n=2 ➔ $h[2] = a_0\delta[2] + a_1\delta[1] + a_2\delta[0] + a_3\delta[-1] + a_4\delta[-2] = a_2$

n=3 ➔ $h[3] = a_0\delta[3] + a_1\delta[2] + a_2\delta[1] + a_3\delta[0] + a_4\delta[-1] = a_3$

n=4 ➔ $h[4] = 0 + 0 + 0 + 0 + a_4\delta[0] = a_4$

n=5 ➔ $h[5] = 0 + 0 + 0 + 0 + a_4\delta[1] = 0$

For n ≥ 5, h[n] = 0, since the system has only four delay elements.

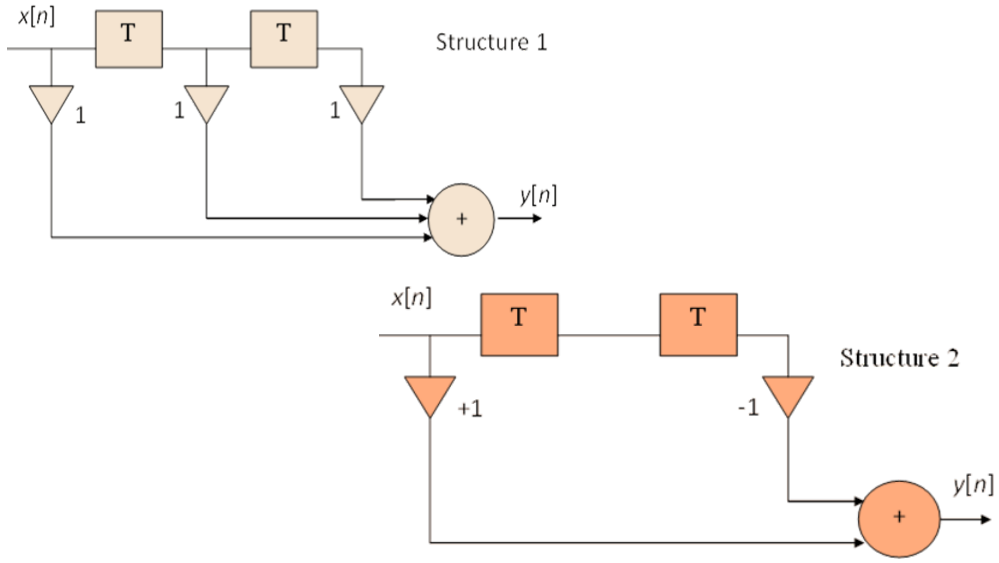


a_0 a_1 a_2 a_3 and a_4 are called coefficients (+ or -) or constants.

$$h[n] = \{a_0, a_1, a_2, a_3, a_4\}, \qquad h[n] = 0, n \geq 5$$

Example 2

- Two FIR systems (Structures) are shown below:
  (a) Write the difference equation;
  (b) Calculate the impulse response.



Structure 1:

Structure 2:

$$y[n] = x[n] - x[n-2]$$

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems

- Discrete-Time Systems (Ref pp. 55-61)
- Classification of Discrete-Time Systems (Ref pp. 61-69)
- Linear Time Invariant (LTI) Systems (Ref pp. 71-100)
- Introduction to z-Transform (Ref pp. 151-174)
- Pole-Zero Description of Discrete-Time Systems (Ref pp. 174-183)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (Ref pp. 247-297)

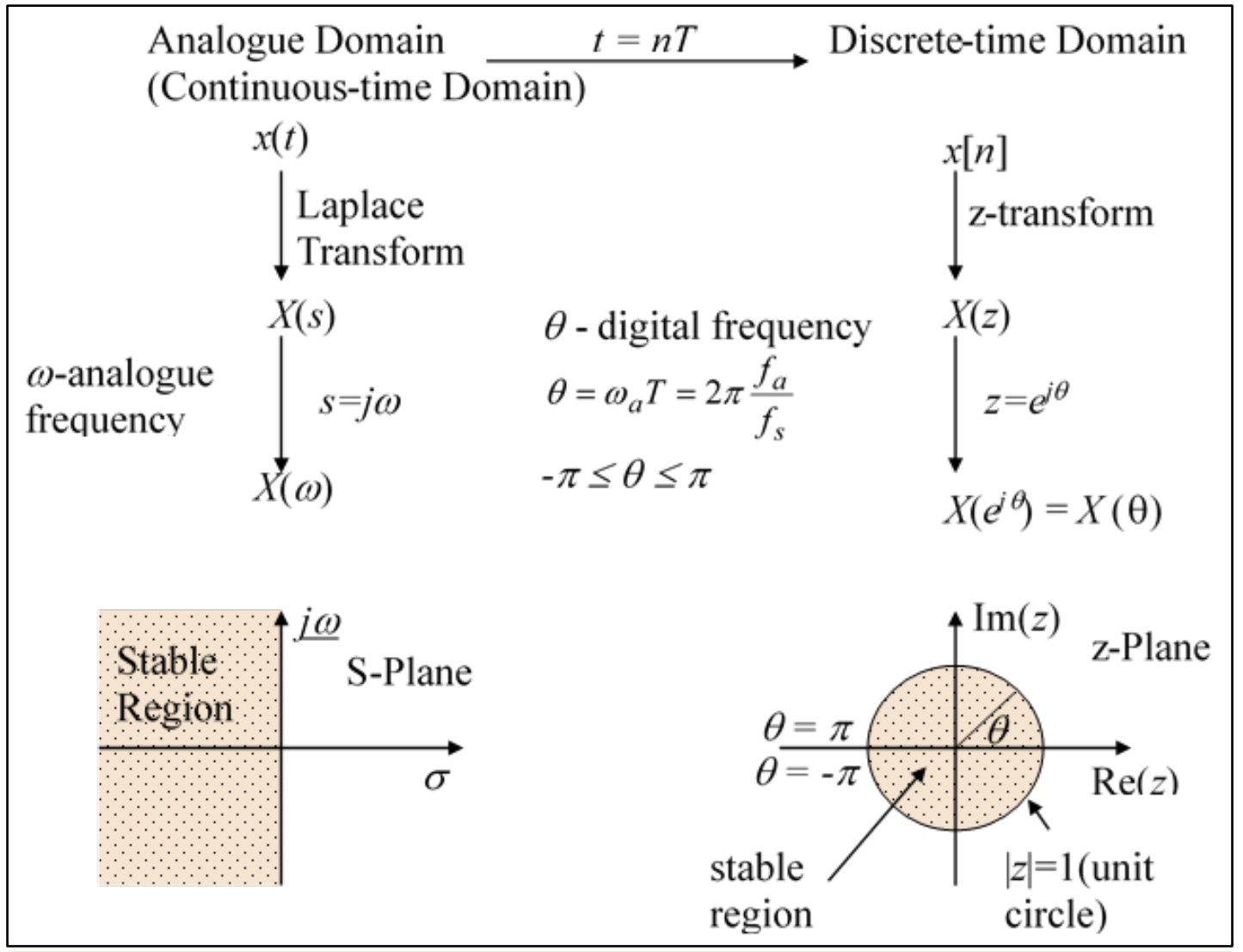
Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

z-Transform

- The Laplace transform is widely used for the analysis of continuous-time LTI systems described by differential equations.
- Similarly, the z-transform is used to analyse discrete-time LTI systems described by difference equations.
- The z-transform allows us to study system characteristics, stability and frequency response, and is also useful for developing structures for implementing discrete-time systems.
- We concentrate on the operational properties of the z-transform and its applications in system Analysis.



The jω-axis in the s-plane corresponds to the unit circle in the z-plane when we study frequency response.

z-Transform

- The **two-sided z-transform** of a discrete-time signal is given by :

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

where *z* is a complex variable; $z = re^{j\theta}$, where r is the magnitude and θ is the angle.

On the unit circle (r=1), $z = e^{j\theta}$ and θ = digital frequency;

- For a causal sequence, $x[n] = 0$ for $n < 0$. Therefore, its **z-transform** can be written as :

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

- Clearly, the z-transform is a power series with an infinite number of terms and so may not converge for all values of *z*.

- The region where the z-transform converges is known as the **region of convergence (ROC)** and in this region the values of *X*(*z*) are finite.

- **The step sequence:** $x[n] = u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$

$$X(z) = \sum_{n=0}^{\infty} 1 \cdot z^{-n} = 1 + z^{-1} + z^{-2} + \cdots = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

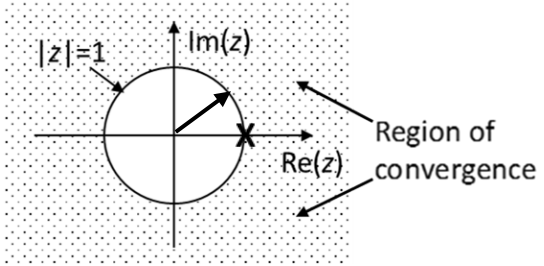
This is a geometric series with a common ratio of z^{-1}. The series converges if $|z^{-1}| < 1$ or equivalently if $|z| > 1$ and diverges if $|z| < 1$.

Let z =2 , $X(z) = 1 + 2^{-1} + 2^{-2} + \ldots = \frac{1}{1-2^{-1}} = 2$

Let z = 0.5, $X(z) = 1 + \left(\frac{1}{2}\right)^{-1} + \left(\frac{1}{2}\right)^{-2} + \cdots = 1 + 2 + 4 + \cdots$

- In this case, the z-transform is valid everywhere outside a circle of unit radius whose centre is at the origin (see below):

$|z|=1$ is a circle of unit radius referred to as the 'unit circle'



- The Region of Convergence (ROC) is $|z| > 1$. Its boundary is the circle $|z|=1$, which passes through the **pole** at z=1.

z-Transform

The delta sequence: $\delta[n]$

$$Z\{\delta[n]\} = \sum_{n=0}^{\infty} \delta[n] z^{-n} = 1$$

The geometric sequence: $x[n] = a^n$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-\frac{a}{z}} = \frac{z}{z-a} \qquad \left|\frac{a}{z}\right| < 1$$

or equivalently |z| > |a|

when $a = 1$, $x[n] = 1$ for $n \geq 0$ ie. $x[n] = u[n]$

$$X(z) = \frac{z}{z-1} \qquad \text{ROC: } |z|>1$$

The complex exponential sequence: $x[n] = e^{j\theta n}u[n]$

$$Z\{e^{j\theta n}u[n]\} = \sum_{n=0}^{\infty} e^{j\theta n} z^{-n} = \frac{1}{1-\frac{e^{j\theta}}{z}} = \frac{z}{z-e^{j\theta}}$$

$$\therefore Z\{\cos n\theta\} = \frac{z(z-\cos\theta)}{z^2 - 2z\cos\theta + 1}$$

$$\therefore Z\{\sin n\theta\} = \frac{z\sin\theta}{z^2 - 2z\cos\theta + 1}$$

Properties of z-Transform

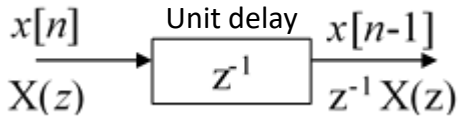
Linearity: $ax[n] + by[n] \overset{Z}{\longleftrightarrow} aX(z) + bY(z)$

Shifting property (Delay theorem):

$$x[n-k] \overset{Z}{\longleftrightarrow} z^{-k}X(z)$$

- A very important property of the z-transform is the delay theorem.
- Example: (1) $Z\{x[n-1]\} = z^{-1}X(z)$

  (2) $Z\{x[n-2]\} = z^{-2}X(z)$



Time reversal : $x[-n] \overset{Z}{\longleftrightarrow} X\left(\frac{1}{z}\right) = X(z^{-1})$

Example: Find the z-transform of $x[n] = u[-n]$.

$$u[n] \overset{Z}{\longleftrightarrow} \frac{z}{z-1}; \quad u[-n] \overset{Z}{\longleftrightarrow} \frac{z^{-1}}{z^{-1}-1} = \frac{1}{1-z} \qquad \text{ROC: } |z| < 1$$

OR $u[-n] = 1, \quad n \leq 0,$

$$\sum_{n=-\infty}^{0} z^{-n} = 1 + z + z^2 + \cdots = \frac{1}{1-z} \quad |z| < 1$$

z-Transform

Properties of z-Transform

Multiplication by exponential sequence:

$$a^n x[n] \overset{z}{\longleftrightarrow} X(a^{-1}z)$$

As a special case if $x[n]$ is multiplied by $e^{jn\theta}$

$$e^{j\theta n} x[n] \overset{z}{\longleftrightarrow} X(e^{-j\theta}z)$$

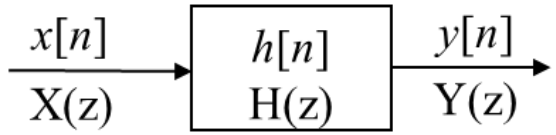
Differentiation in the z-domain:

$$nx[n] \overset{z}{\longleftrightarrow} -z\frac{d}{dz}X(z)$$

Discreate convolution:

If $y[n] = x[n] * h[n]$

then $Y(z) = X(z) \cdot H(z)$



- **Convolution** in the time domain corresponds to multiplication in the z-domain.

$$x[n] * h[n] \overset{z}{\longleftrightarrow} X(z).H(z)$$

Example 3

- **Discrete convolution in z-domain:** Compute the convolution $y[n]$ of the discrete-time signals given by (assume that both sequences start at n=0]:

  $x_1[n] = [1, -2, 1]$; $\{x_1[0] = 1, x_1[1] = -2, x_1[2] = 1\}$

  $x_2[n] = 1$ for $0 \le n \le 5$,
  $x_2[n] = 0$ elsewhere

Ans: $y[n] = x_1[n] * x_2[n] \Rightarrow Y(z) = X_1(z) \cdot X_2(z)$

$X_1(z) = 1 - 2z^{-1} + z^{-2}$ or $X_1(z) = (1 - z^{-1})^2$

$X_2(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} = \frac{1-z^{-6}}{1-z^{-1}}$

$Y(z) = X_1(z) \cdot X_2(z) = (1 - z^{-1})^2 \frac{1-z^{-6}}{1-z^{-1}} = (1 - z^{-1})(1 - z^{-6})$

$= 1 - z^{-1} - z^{-6} + z^{-7}$

↓ Inverse z-transform

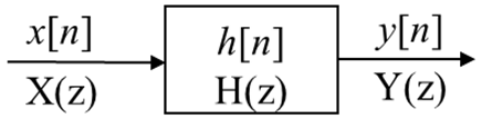
$y[n] = [1, -1, 0, 0, 0, 0, -1, 1]$

Note: Length of $y[n]$ = Length of $x_1[n]$ + Length of $x_2[n]$ - 1
[8 samples = 3 samples + 6 samples - 1]

z-Transform

Example 4

- Convolution of a Finite-Length Sequences: The convolution of two finite-length sequences is also finite in length. Find the convolution of $h[n]$ = [2, 5, 0, 4] and $x[n]$ = [4, 1, 3];



If $y[n] = x[n] * h[n]$
then $Y(z) = X(z) \cdot H(z)$

Method 1: Convolution using z-transform

$Y(z) = X(z) \cdot H(z)$

$X(z) = 4 + 1\,z^{-1} + 3z^{-2}$

$H(z) = 2 + 5z^{-1} + 4\,z^{-3}$

$Y(z) = 8 + 22z^{-1} + 11z^{-2} + 31z^{-3} + 4z^{-4} + 12z^{-5}$

↓ Inverse z-transform

$y[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$

Length of $y[n]$ = Length of $x[n]$ + Length of $h[n]$ –1 = 3 + 4 – 1= 6

Method 2: Convolution in Discrete-time domain:

$$y[n] = x[n] * h[n] = \sum_{i=-\infty}^{\infty} x[i]\,h[n-i]$$

- The flipped sequence is $h[-i]$ = [4, 0, 5, 2]. Line up flipped sequence below $x[n]$ to begin overlap and shift it successively, summing the product sequence to obtain the discrete convolution

0 shift

$x[n]$: 4 1 3
4 0 5 2 ← $h[-i]$
$y[0]$ = Sum of products = 8

1 shift

$x[n]$: 4 1 3
4 0 5 2 ← $h[1-i]$
$y[1] = (5x4)+(2x1) = 22$

2 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[2-i]$
$y[2] = 0 + 5 + 6 = 11$

3 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[3-i]$
$y[3] = 16 + 0 + 15 = 31$

4 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[4-i]$
$y[4] = 4 + 0 = 4$

5 shifts

$x[n]$: 4 1 3
4 0 5 2 ← $h[5-i]$
$y[5] = 12$

$\therefore y[n] = x[n] * h[n] = [8 \quad 22 \quad 11 \quad 31 \quad 4 \quad 12]$

z-Transform

Transfer Function definition

- For an LTI system with zero initial conditions, the transfer function is defined as the ratio of the output *Y(z)* and the input *X(z)* or as the z-transform of the system impulse response *h[n]*.
- Given the following difference equation, find the transfer function $H(z) = \frac{Y(z)}{X(z)}$.

$$y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] + b_1y[n-1] + b_2y[n-2]$$

Take the z-transform of both sides:

$$Y(z) = a_oX(z) + a_1z^{-1}\,\mathrm{X(z)} + a_2z^{-2}\,\mathrm{X(z)} + b_1z^{-1}\,Y(z) + b_2z^{-2}\,Y(z)$$

$$Y(z) - b_1\,z^{-1}\,Y(z) - b_2\,z^{-2}\,Y(z) = a_o\,X(z) + a_1\,z^{-1}\,\mathrm{X(z)} + a_2\,z^{-2}\,\mathrm{X(z)}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_o + a_1\,z^{-1} + a_2\,z^{-2}}{1 - b_1\,z^{-1} - b_2\,z^{-2}}$$

- The transfer function is thus a ratio of polynomials in z^{-1}.
- A LTI system may be described by its **transfer function**, **impulse response**, **difference equation** or its **poles** (roots of the denominator) and **zeros** (roots of the numerator)
- Given one form, it is possible to obtain any of the other forms.

Example 5

- Find the difference-equation of the following transfer function: $H(z) = \frac{5z+2}{z^2+3z+2}$

Step1: First rewrite *H*(*z*) as a ratio of polynomials in z^{-1}

$$H(z) = \frac{Y(z)}{X(z)} = \frac{5z^{-1} + 2z^{-2}}{1 + 3z^{-1} + 2z^{-2}}$$

$$Y(z) + 3\,Y(z)z^{-1} + 2\,Y(z)z^{-2} = 5\,\mathrm{X(z)}\,z^{-1} + 2\mathrm{X(z)}\,z^{-2}$$

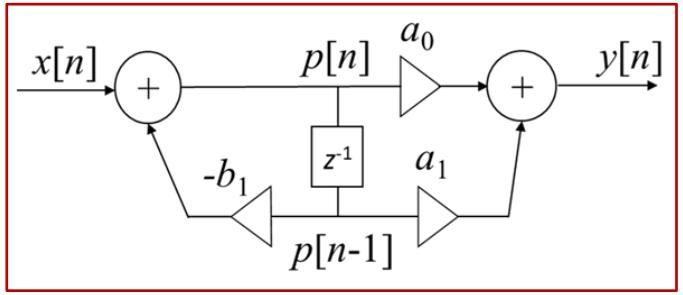
Step 2: Take inverse z-transform

$$y[n] + 3y[n-1] + 2y[n-2] = 5x[n-1] + 2x[n-2]$$

$$y[n] = 5x[n-1] + 2x[n-2] - 3y[n-1] - 2y[n-2]$$

Example 6

Determine the Transfer Function $H(z) = \frac{Y(z)}{X(z)}$.



$$\frac{P(z)}{X(z)} = \frac{1}{1 + b_1z^{-1}}$$

$$\frac{Y(z)}{P(z)} = a_0 + a_1z^{-1}$$

$$\frac{Y(z)}{X(z)} = \frac{a_0 + a_1z^{-1}}{1 + b_1z^{-1}}$$

z-Transform Table

Basic z-Transform

Signal $x[n]$	z-Transform $X(z)$	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|\alpha\|$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|\alpha\|$
$\cos(n\theta)u[n]$	$\frac{1-z^{-1}\cos\theta}{1-2z^{-1}\cos\theta+z^{-2}}$	$\|z\| > 1$
$\sin(n\theta)u[n]$	$\frac{z^{-1}\sin\theta}{1-2z^{-1}\cos\theta+z^{-2}}$	$\|z\| > 1$
$r^n\cos(n\theta)u[n]$	$\frac{1-rz^{-1}\cos\theta}{1-2rz^{-1}\cos\theta+r^2z^{-2}}$	$\|z\| > \|r\|$
$r^n\sin(n\theta)u[n]$	$\frac{rz^{-1}\sin\theta}{1-2rz^{-1}\cos\theta+r^2z^{-2}}$	$\|z\| > \|r\|$

z-Transform Properties

Property	Signal	z-Transform
Linearity	$ax[n]+by[n]$	$aX(z)+bY(z)$
Time shifting (delay)	$x[n-k]$	$z^{-k}X(z)$
Multiplication by exponential	$a^n x[n]$	$X(a^{-1}z)$
Time reversal	$x[-n]$	$X(z^{-1})$
Convolution	$x[n]*y[n]$	$X(z)Y(z)$
Multiplication by n	$nx[n]$	$-z\frac{dX(z)}{dz}$

$$S_N = 1 + z^{-1} + z^{-2} + \cdots + z^{-(N-1)} = \frac{1-z^{-N}}{1-z^{-1}}$$

as $N \to \infty$,

$$S_\infty = \frac{1}{1-z^{-1}}, \qquad |z| > 1.$$

Inverse z-Transform

Partial Fraction Method

- The inverse z-transform allows us to recover the discrete-time sequence. The inverse z-transform depends on both $X(z)$ and its region of convergence (ROC).

$$x[n] = \mathbf{Z}^{-1}[X(z)]$$

where $X(z)$ is the z-transform of $x[n]$.

Example 7

- Find the causal sequence $x[n]$ corresponding to the following z-transform:

$$X(z) = \frac{z^{-1}}{1-0.25z^{-1}-0.375z^{-2}}$$

Answer:

$$X(z) = \frac{z}{(z-0.75)(z+0.5)} = z\left[\frac{A}{z-0.75} + \frac{B}{z+0.5}\right] = z\left[\frac{\left(\frac{4}{5}\right)}{z-0.75} + \frac{\left(-\frac{4}{5}\right)}{z+0.5}\right]$$

$$X(z) = \left[\frac{\left(\frac{4}{5}\right)}{1-0.75z^{-1}} - \frac{\left(\frac{4}{5}\right)}{1+0.5z^{-1}}\right]$$

Inverse z-transform

$$x[n] = \frac{4}{5}\left[(0.75)^n - (-0.5)^n\right] u[n] \quad ; \textbf{ROC } |z| > 0.75$$

Example 8

- Find h[n] for the following system and show that the system is stable if $|a|<1$:

$$H(z) = \frac{1}{1+az^{-1}} - \frac{az^{-1}}{1+az^{-1}}$$

Inverse z-transform

Answer:

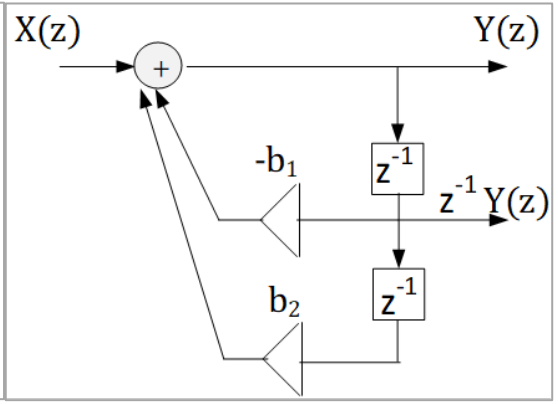
$$h[n] = \{(-a)^n u[n]\} - a\{(-a)^{n-1} u[n-1]\}$$

Since $|a| < 1$, $(-a)^n$ and $(-a)^{n-1}$ are decaying functions. Therefore, $h[n]$ is a decaying function. Hence, the system is stable.

Example 9

- Draw a structure for the following Transfer function:

$$H(z) = \frac{z^{-1}}{1+b_1z^{-1}-b_2z^{-2}}$$



ELEC3104: Digital Signal Processing

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Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Pole-Zero Description of Discrete Time Systems

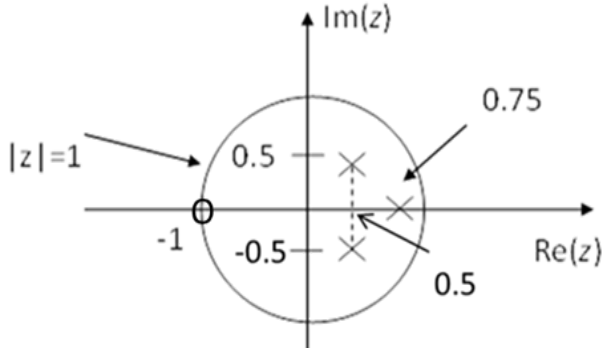
Poles and Zeros

- The **zeros** of the transfer function *H(z)* are the values of *z* for which *H(z)=0*. The **poles** of $H(z)$ are the values of *z* for which the denominator of *H(z) is zero*. If *H(z)* is a rational function , then

$$H(z) = \frac{Y(z)}{X(z)} = \frac{a_0 + a_1 z^{-1} + \ldots + a_M z^{-M}}{1 + b_1 z^{-1} + \ldots\ldots + b_L z^{-L}}$$

$$= a_0\, z^{L-M}\, \frac{(z - z_1)(z - z_2)\ldots\ldots(z - z_M)}{(z - p_1)(z - p_2)\ldots\ldots(z - p_L)}$$

- The complex quantities (or may be real) z_1, z_2, z_3 …. are called zeros of *H(z)* and the complex quantities (or may be real) p_1, p_2, p_3 … are called the poles of *H(z)*.
- We thus see that *H(z)* is completely determined , except for the constant a_0, by the values of poles and zeros.
- The information contained in the z-transform can be conveniently displayed as a poles-zero diagram (see figure on the right)
- In the diagram, 'X' marks the position of a pole and 'O' denotes the position of a zero

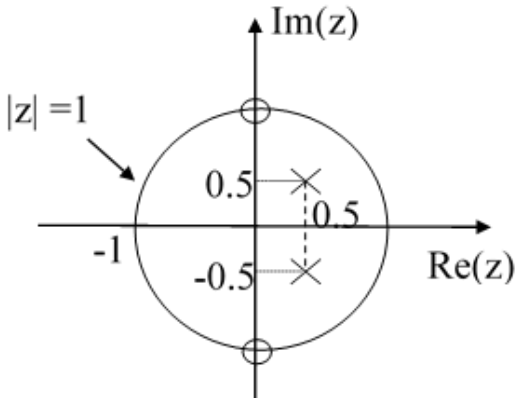


- The poles are located at $z = 0.5 \pm 0.5j$ and $z = 0.75$; a single zero is at $z = -1$.
- An important feature of the pole-zero diagram is the unit circle $|z|=1$.
- The pole-zero diagram provides insight into the properties of a discrete-time system.
- From the locations of the poles and zeros we can infer important characteristics of the system, including its frequency response and stability.
- **For a causal LTI system to be stable, all poles must lie strictly inside the unit circle. Zeros may lie inside, on, or outside the unit circle.**

Pole-Zero Description of Discrete Time Systems

Example 10

- Determine the transfer function *H(z)* of a discrete-time system for the pole-zero diagram shown below:

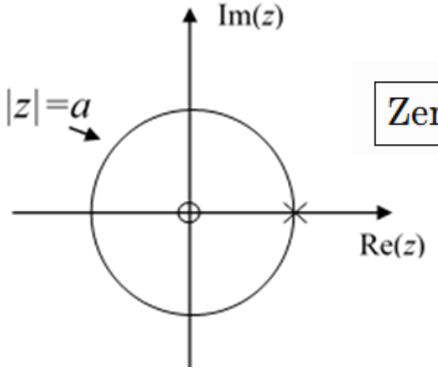


$$H(z) = \frac{K(z-j1)(z+j1)}{(z-0.5-0.5j)(z-0.5+0.5j)}$$

$$H(z) = \frac{K(1+z^{-2})}{1-z^{-1}+0.5z^{-2}}$$

Example 11

- Determine the pole-zero plot for $H(z) = \frac{z}{z-a}$



Zero: $z = 0$, Pole: $z = a$

ROC: $|z| > |a|$

Example 12

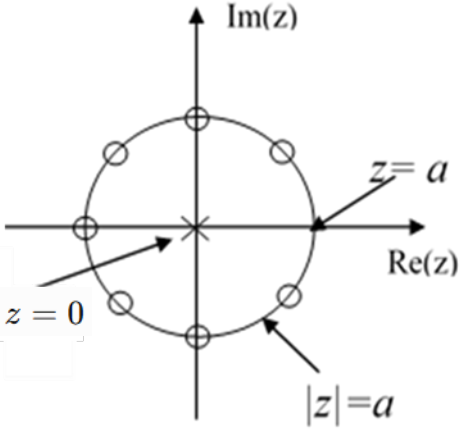
- Determine the pole-zero plot for

$$H(z) = \frac{z^M - a^M}{z^{M-1}(z-a)}$$

$$z^M - a^M = (z-a)\left(z^{M-1} + az^{M-2} + \cdots + a^{M-1}\right)$$

$$= \frac{(z-z_1)(z-z_2).....(z-z_{(M-1)})}{z^{M-1}}$$

- The zero z = *a* cancels the pole at z = *a*. Thus, *H(z)* has *M*-1 zeros and *M*-1 poles as shown in the diagram below for *M* = 8.

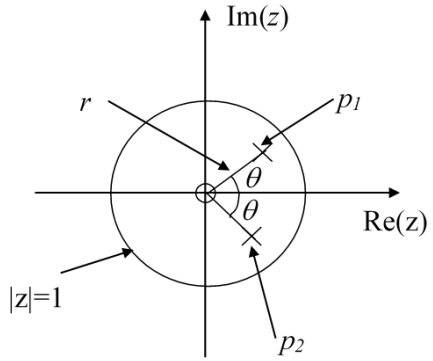


7 coincident poles at $z = 0$

Pole-Zero Description of Discrete Time Systems

Time domain behaviour of a system with a pair of complex conjugate poles

- Consider a system, *H(z)* with two complex conjugate poles in the z-plane:



$$p_1 = re^{j\theta}$$
$$p_2 = re^{-j\theta}$$
$$z_1 = 0 \quad (zero)$$

} Poles

- The transfer function corresponding to the pole-zero pattern:

$$H(z) = \frac{z}{(z-re^{j\theta})(z-re^{-j\theta})} = z\left[\frac{A}{z-re^{j\theta}} + \frac{B}{z-re^{-j\theta}}\right]$$

$$= z\left[\frac{\frac{1}{2jr\sin\theta}}{z-re^{j\theta}} - \frac{\frac{1}{2jr\sin\theta}}{z-re^{-j\theta}}\right] = \frac{1}{j(2r\sin\theta)}\left[\frac{1}{1-re^{j\theta}z^{-1}} - \frac{1}{1-re^{-j\theta}z^{-1}}\right]$$

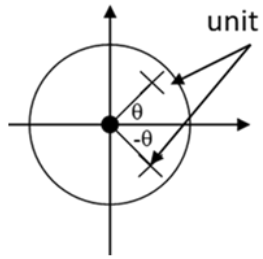
$$h(n) = \frac{1}{j2r\sin\theta}\left[(re^{j\theta})^n - (re^{-j\theta})^n\right] = \frac{r^n}{j2r\sin\theta}\left[e^{j\theta n} - e^{-j\theta n}\right]$$

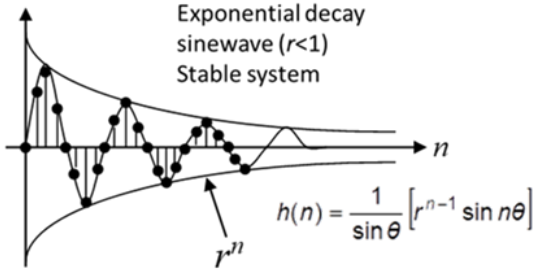
$$\boldsymbol{h(n) = \frac{1}{\sin\theta}\left[r^{n-1}\sin n\theta\right]}$$,← θ = frequency of oscillation

- This is the impulse response of the 2nd order system with complex poles. We note that the impulse response will decay away to zero provided r is less than one [$r < 1$].

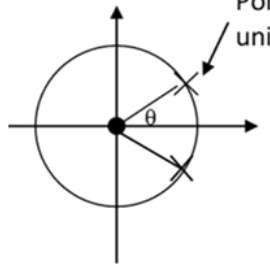
- Recall that *r* is also the distance from the origin in the z-plane to the poles p_1 or p_2, so that system will be stable if the poles in the z-plane lie inside the unit circle.

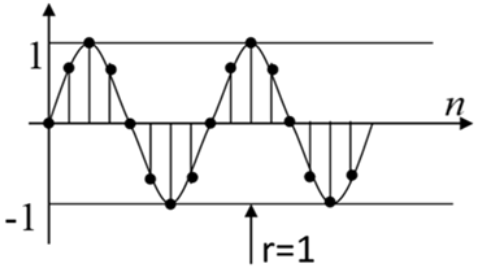
Case1:



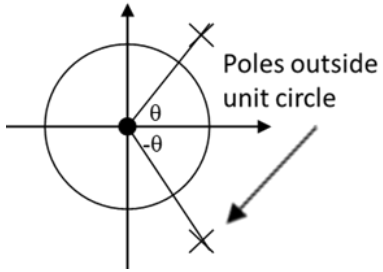


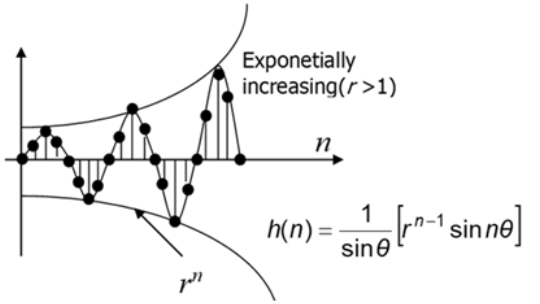
Case 2:





Case3:





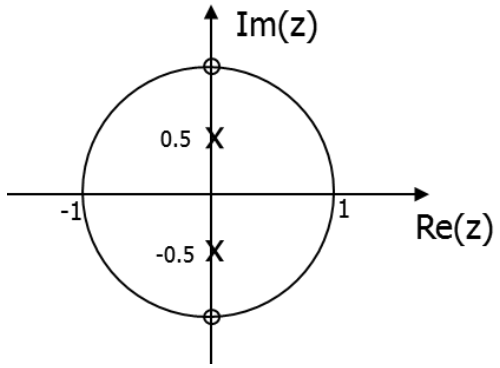
Pole-Zero Description of Discrete Time Systems

Stability Criteria

- A system that is both **stable and causal** must have all its poles inside the unit circle in the z-plane.
- We cannot have a pole outside the unit circle, since the inverse transform of a pole located outside the unit circle will contribute either a right-sided increasing exponential term, which is not stable, or a left-sided decaying exponential term, which is not causal.

Example 13

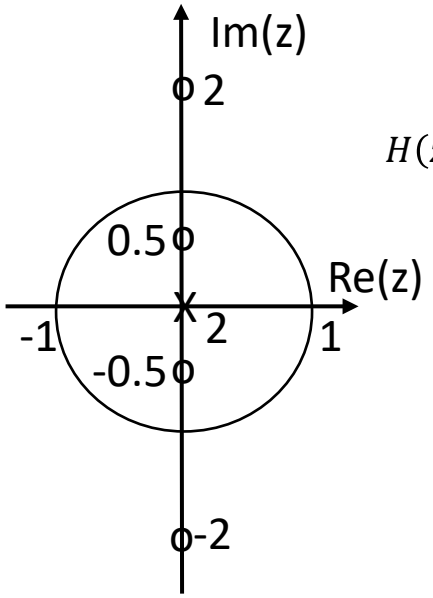
- Determine the transfer function H(z) of a discrete-time system with the pole-zero map given below:



$$H(z) = K\frac{(z-j)(z+j)}{(z-0.5j)(z+0.5j)} = K\frac{(z^2+1)}{(z^2+0.25)}$$

Example 14

- What is the z-transform corresponding to the pole-zero pattern shown in the diagram below. Does it represent a symmetrical signal?



$$H(z) = K\frac{(z-j0.5)(z+j0.5)(z-j2)(z+j2)}{z^2}$$

$$H(z) = K\frac{(z^4+4.25z^2+1)}{z^2}$$

$$H(z) = K\,(z^2+4.25+z^{-2})$$

$$h[n] = K\,\{1, 0, 4.25, 0, 1\}$$

n = -2 n = 2

h[n] has even symmetry

$h[n] = h[-n]$

- Checking for symmetry , we find that $H(z) = H(z^{-1})$ and thus h[n] is even symmetric.
- An even-symmetric sequence is generally noncausal, since it has values for both positive and negative n.

ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems

- Discrete-Time Systems (Ref pp. 55-61)
- Classification of Discrete-Time Systems (Ref pp. 61-69)
- Linear Time Invariant (LTI) Systems (Ref pp. 71-100)
- Introduction to z-Transform (Ref pp. 151-174)
- Pole-Zero Description of Discrete-Time Systems (Ref pp. 174-183)
- Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT) (Ref pp. 247-297)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

DTFT

- The Fourier Transform is a very important tool in many aspects of engineering.
- In the case of digital signal processing, it allows signals to be represented in terms of their constituent frequency components, i.e. as signals in the 'frequency domain'.
- In some cases, signal processing operations can be performed in the frequency domain rather than in the time domain, which can have a number of advantages.
- Before proceeding, it is worth noting that in signal processing what is referred to as 'the Fourier transform' may in fact be one of four possible types of Fourier transform, illustrated in the adjacent Table.
- The difference between each type of transform is whether the time domain signal is continuous – time or discrete-time and whether its frequency domain representation is continuous (i.e. a function of real numbers ω or θ) or discrete (i.e. a function of integers *n* or *k*)

	Time domain	Frequency domain
DTFT	Discrete $x[n]$	Continuous & periodic $X(\theta)$
DFT	Finite discrete sequence $x[n]$	Discrete $X[k]$

- The Fourier transform of a discrete-time signal is called the Discrete-Time Fourier Transform (DTFT).
- For discrete-time signals, the Fourier transform pair is defined by:

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \quad x[n] \xleftrightarrow{\text{DTFT}} X(\theta) \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\, d\theta$$

- where θ is the digital frequency or relative frequency ($-\pi \le \theta \le \pi$) and $\theta = \omega T$. $X(\theta)$ is periodic with period 2π. (ie. $X(\theta + 2\pi) = X(\theta)$). Note that although $x[n]$ is discrete, ***X*(θ) is a continuous function** of frequency θ (hence the integral in the inverse transform).
- Note: Unlike the DTFT, the Fourier transform of a continuous-time signal is not periodic in frequency.

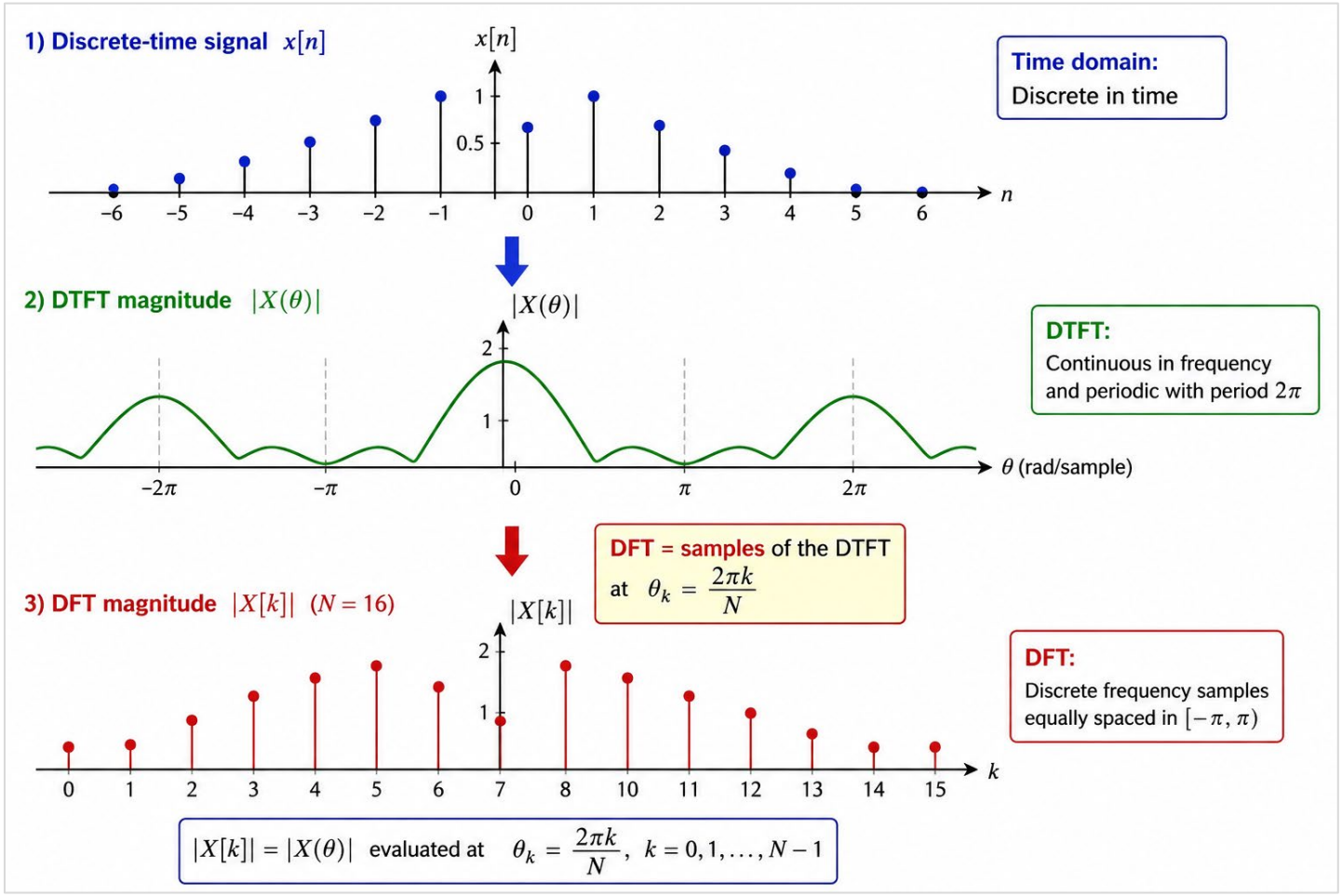
Discrete Time Fourier Transform (DTFT) and Discrete Fourier Transform (DFT)

Discrete-time signal $x[n]$ $\xrightarrow{\text{DTFT}}$ Continuous and periodic frequency spectrum $X(\theta)$

DTFT: continuous in frequency and periodic with period 2π.

DFT: discrete frequency samples of the DTFT.

DFT = equally spaced frequency samples of the DTFT



	Time domain	Frequency domain
DTFT	Discrete $x[n]$	Continuous & periodic $X(\theta)$
DFT	Finite discrete sequence $x[n]$	Discrete $X[k]$

$$X[k] = X(\theta)\Big|_{\theta=\frac{2\pi k}{N}}, \qquad k = 0, 1, \ldots, N-1$$

Discrete Time Fourier Transform (DTFT)

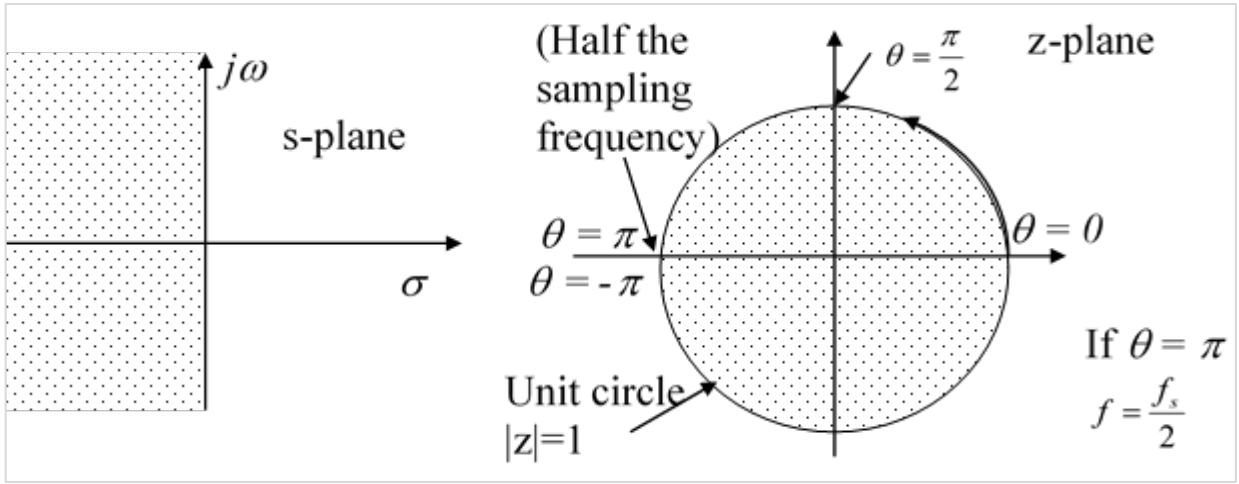
Relationship between the z-transform and Laplace transform

- If we let $z = e^{sT}$, then $z = e^{(\sigma + j\omega)T}$ (where T is the sampling period)

$$z = e^{\sigma T} \cdot e^{j\omega T} = e^{\sigma T} \cdot e^{j\theta} \qquad -\pi \le \theta \le \pi$$

$$\therefore |z| = e^{\sigma T} \text{ and } \arg(z) = \omega T = 2\pi \frac{f}{f_s} = \theta \quad (\theta = \text{digital frequency})$$

- As ω varies from -∞ to ∞ the s-plane is mapped to the z-plane as shown in figure below:



Mapping of the s-plane to the z-plane.

- The entire $j\omega$ axis $(\sigma = 0)$ in the s-plane is mapped onto the unit circle $(\sigma = 0;\ |z| = e^{\sigma T} = 1)$.
- The left-hand s-plane $(\sigma < 0)$ is mapped to the inside of the unit circle $(|z| < 1)$.
- The right-hand s-plane $(\sigma > 0)$ is mapped to the outside of the unit circle $(|z| > 1)$.

Frequency Axis and the Unit Circle

- In terms of frequency response, the $j\omega$ axis is the most important in the s-plane. In this case $\sigma = 0$ and the frequency points in the s-plane are related to points on the z-plane unit circle by:

$$z = e^{\sigma T}\, e^{j\omega T} = 1 \cdot e^{j\theta} = e^{j\theta}$$

$$\boldsymbol{z = e^{j\theta}}$$

Thus the mapping $s = j\omega \longleftrightarrow z = e^{j\theta}$ establishes the direct connection between continuous frequency ω (rad/s) and digital frequency θ (rad/sample), where $\theta = \omega T$ and $-\pi \le \theta \le \pi$.

Discrete Time Fourier Transform (DTFT)

Frequency Response Estimation

- There are many instances when it is necessary to evaluate the frequency response of discrete-time systems.
- The frequency response of a system can be readily obtained from its z-transforms.
- For example, if we set $z = e^{j\theta}$, (i.e evaluate the z-transform around the unit circle) we obtain the Discrete Time Fourier Transform (DTFT) of the system.

$$H(\theta) = H(z)|_{z=e^{j\theta}}, \quad -\pi \leq \theta \leq \pi$$

If $H(z)$ is the z-transform of $h[n]$, then
$H(\theta)$ is the DTFT of $h[n]$ and is periodic in θ with period 2π.

- **Key Idea:** The DTFT is the z-transform evaluated on the unit circle ($z = e^{j\theta}$) . This gives the frequency response over all digital frequencies.

Discrete Time Fourier Transform (DTFT)

Example 15

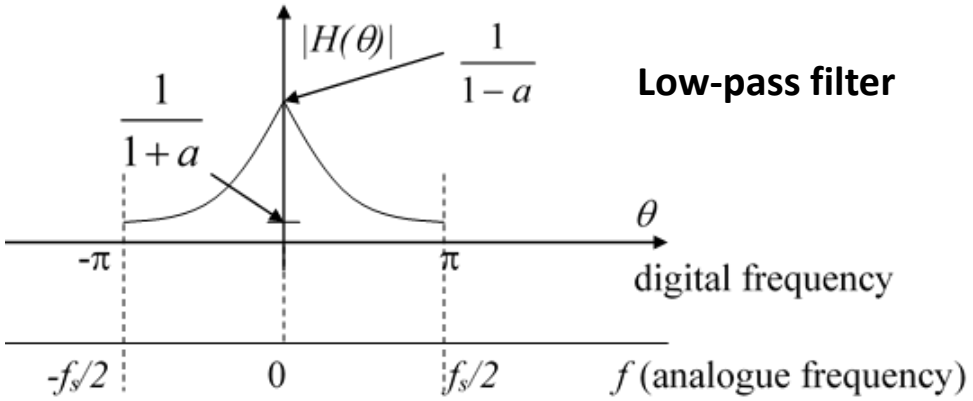
- Determine and sketch the magnitude response for the following system:

$$H(z) = \frac{1}{1-az^{-1}} \qquad 0 < a < 1$$

- To find $H(\theta)$ (the frequency response) , substitute $z = e^{j\theta}$

$$H(\theta) = H(z)\big|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}} = \frac{1}{(1-a\cos\theta)+ja\sin\theta}$$

$$|H(\theta)| = \frac{1}{\sqrt{(1-a\cos\theta)^2+(a\sin\theta)^2}} = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$



Low-pass filter

- **DC gain : $|H(0)| = {}^{1}/_{1-a}$** {when $\theta = 0$}

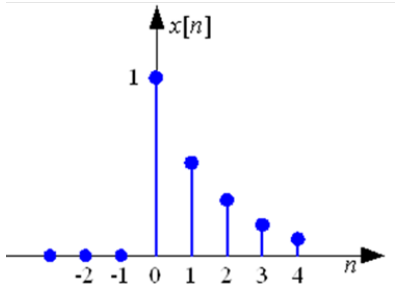
Example 16

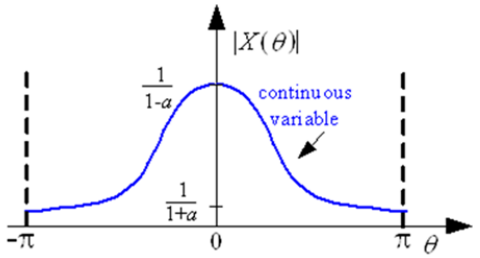
- Determine DTFT of the following sequence and sketch its magnitude response.

$$x[n] = \begin{cases} 0 & n < 0 \\ a^n & n \geq 0, |a| < 1 \end{cases} \qquad X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}$$

$$X(\theta) = \sum_{n=0}^{\infty} a^n e^{-jn\theta} = \sum_{n=0}^{\infty} (ae^{-j\theta})^n = \frac{1}{1-ae^{-j\theta}}$$

$$X(\theta) = \frac{1}{(1-a\cos\theta)+ja(\sin\theta)} \qquad \therefore |X(\theta)| = \frac{1}{\sqrt{1-2a\cos\theta+a^2}}$$





- ***Note:*** The above $X(\theta)$ can be obtained using the z-transform as well: $x[n] = a^n u[n] \Rightarrow X(z) = \frac{1}{1-az^{-1}}$

$$X(\theta) = X(z)\Big|_{z=e^{j\theta}} = \frac{1}{1-ae^{-j\theta}}$$

Discrete Time Fourier Transform (DTFT)

Example 17

- Consider a system described by $y[n] = x[n] - x[n-1]$. Find (a) the impulse response; (b) transfer function $H(z) = \frac{Y(z)}{X(z)}$; (c) Discrete time Fourier Transform $H(\theta)$. Sketch the magnitude response.

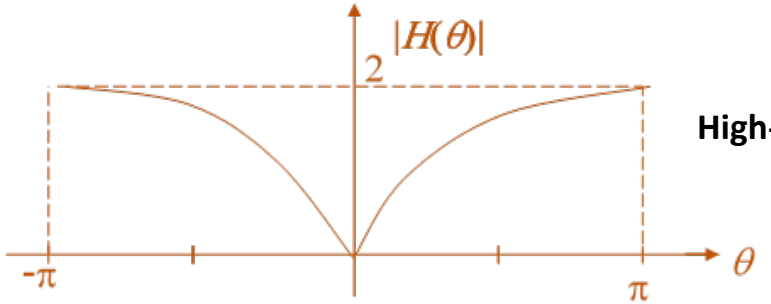
Answer:

$(a)\ h[n] = \{1, -1\}$;

$(b)\ Y(z) = X(z) - z^{-1}X(z);\ H(z) = 1 - z^{-1}$

$(c)\ H(\theta) = 1 - e^{-j\theta}$

$$|H(\theta)| = \sqrt{(1-\cos\theta)^2 + (-\sin\theta)^2} = \sqrt{2 - 2\cos\theta}$$



High-pass filter

- **DC gain : $|H(0)| = 0$** {when $\theta = 0$}

Properties of DTFT

$x[n] \overset{\text{DTFT}}{\rightarrow} X(\theta)$ $x_1[n] \overset{\text{DTFT}}{\rightarrow} X_1(\theta)$ $x_2[n] \overset{\text{DTFT}}{\rightarrow} X_2(\theta)$

- Linearity: $ax_1[n] + bx_2[n] \overset{\text{DTFT}}{\leftrightarrow} aX_1(\theta) + bX_2(\theta)$
- Time shift: $x[n-k] \overset{\text{DTFT}}{\leftrightarrow} e^{-j\theta k}X(\theta)$
- Frequency Shift: $x[n]e^{jn\theta_0} \overset{\text{DTFT}}{\leftrightarrow} X(\theta - \theta_0)$
- Frequency Differentiation: $nx[n] \overset{DTFT}{\leftrightarrow} j\frac{d}{d\theta}X(\theta)$
- Convolution: $x_1[n] * x_2[n] \overset{DTFT}{\leftrightarrow} X_1(\theta)X_2(\theta)$
- Multiplication: $x_1[n]x_2[n] \overset{DTFT}{\leftrightarrow} \frac{1}{2\pi}[X_1(\theta) * X_2(\theta)]$
- Parseval's Theorem:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(\theta)|^2 d\theta$$

Inverse Discrete Time Fourier Transform (IDTFT)

Example 18

- The frequency response of an ideal low pass filter, in the fundamental interval $-\pi \le \theta \le \pi$ is given by

$$H(\theta) = \begin{cases} \frac{1}{2} & 0 \le |\theta| \le \theta_C \\ 0 & \theta_C < |\theta| \le \pi \end{cases}$$

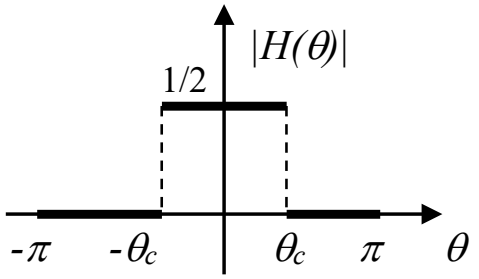
$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta} \quad x[n] \xleftrightarrow{\text{DTFT}} X(\theta) \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

(a) Find the impulse response of the ideal low pass filter;

(b) Sketch the impulse response for $\theta_C = \frac{\pi}{4}$.

Answer:

(a)

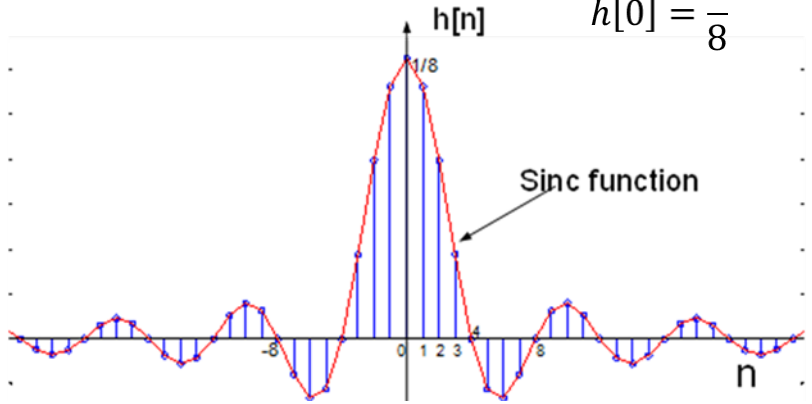


$$h[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\theta)e^{jn\theta}d\theta = \frac{1}{2\pi}\int_{-\theta_C}^{\theta_C}\frac{1}{2}e^{jn\theta}d\theta$$

$$= \frac{1}{4\pi}\frac{e^{j\theta_C n} - e^{-j\theta_C n}}{jn} = \frac{1}{2\pi}\frac{\sin\theta_c n}{n}$$

(b) $\theta_c = \frac{\pi}{4}$ $\quad h(n) = \frac{1}{2\pi}\frac{\sin\frac{n\pi}{4}}{n} = \frac{1}{8}\frac{\sin\frac{n\pi}{4}}{\frac{n\pi}{4}}$

$h[0] = \frac{1}{8}$



Discrete Fourier Transform (DFT)

Discrete Fourier Transform (DFT)

- The importance of the discrete Fourier transform (DFT) for practical applications is hard to overstate.
- If you've ever seen a real time spectrum of just about anything, chances are it was generated using a DFT.
- The DFT allows us to determine the frequency content of a signal, that is, to perform spectral analysis.
- The DFT plays a central role in the implementation of a variety of digital signal processing algorithms, as a result of the existence of the efficient algorithm for the Fast Fourier Transform (FFT).

$$x[n] \xleftrightarrow{DTFT} X(\theta) \quad \text{DTFT}$$

$$x[n] \xleftrightarrow{DFT} X[k] \quad \text{DFT}$$

{DFT mapping $x[n]$ to another sequence}

- The N-point DFT, X[k] is obtained by evaluating the DTFT of an N-sample sequence at N equally spaced frequencies.
- This sampling is conventionally performed at equally spaced frequency points over the period extending over $-\pi \le \theta \le \pi$.

(1) Fourier transform of a discrete signal (DTFT) is

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}, \quad -\pi \le \theta \le \pi$$

(2) $X(\theta) = X(z)|_{z=e^{j\theta}}$ {Evaluate z-transform on the unit circle}

(3) Discrete Fourier Transform (DFT) or N-point DFT

DFT: $X[k] = \sum_{n=0}^{N-1} x[n]e^{-jn\left(\frac{2\pi k}{N}\right)}$ $\boxed{X[k] = X(\theta)\big|_{\theta=2\pi k/N}}$

IDFT: $x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{jn\left(\frac{2\pi k}{N}\right)}$

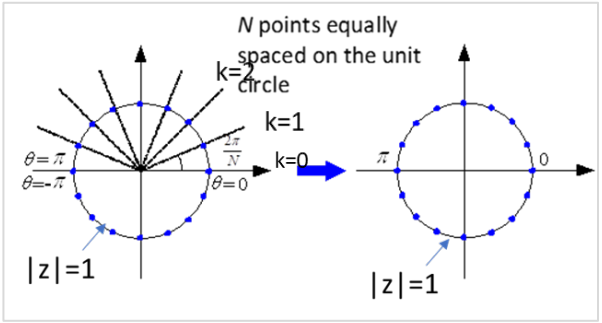
- i.e. sampling $X(\theta)$ at N equally spaced frequency points , k = 0, 1, 2, … N-1. An important property of the DFT is its implied periodicity. $\boxed{X[k+N] = X[k]}$

Number of time samples = N
Number of frequency samples (k) = N

Approximating the DTFT by the DFT

Sampling the unit circle at N points equally spaced in angle:

- The DFT corresponds to sampling the z-transform *X(z)* at *N* equally spaced points around the unit circle ($\theta = \frac{2\pi k}{N}$; $(k = 0, 1, 2, \ldots N-1)$

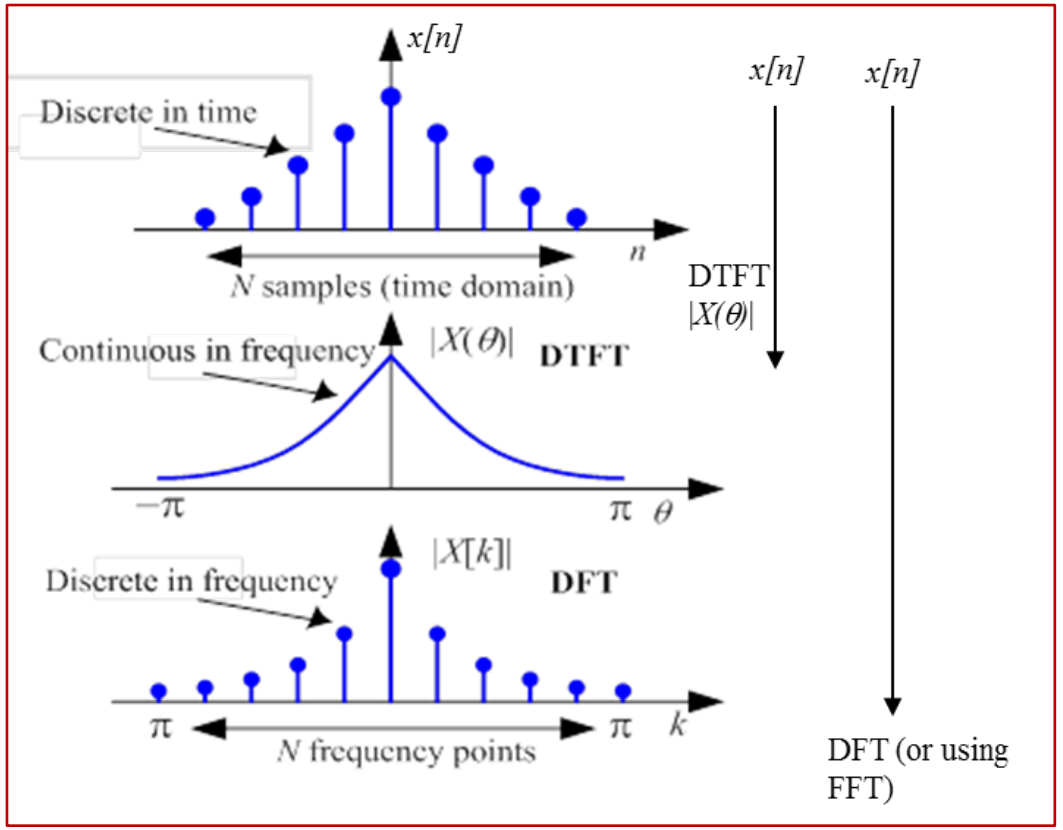


- The diagram on the right shows an example of a sequence $x[n]$ and its corresponding DTFT, X(θ), and the sampled version of DTFT known as DFT, X[k].

Approximating the DTFT by the DFT

- The DFT treats the N-sample data record as one period of a periodic sequence with period N. For a general measured signal, the N-sample data may not contain an integer number of periods.
- For a sinusoid, the data record repeats smoothly only if N contains an integer number of periods of the sinusoid.
- Truncating the signal to N samples is equivalent to applying a finite-length window and may cause **spectral leakage** in the frequency-domain representation.

If x[n] is a finite length sequence of N samples $n = 0, 1, 2 \ldots . N-1$, the DFT gives exact samples of its DTFT $X(\theta)$ at $\theta = \frac{2\pi k}{N}$ $(k = 0, 1, 2, \ldots N-1)$I. The DFT then perfectly recovers the original N samples of $x[n]$.



Number of time samples = *N*
Number of frequency samples (*k*) = *N*

Discrete Fourier Transform (DFT)

DFT calculation

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-jn\frac{2\pi k}{N}} \qquad k = 0,1,2,3,\ldots N\text{-}1$$

$$X[k] = \sum_{n=0}^{N-1} x[n]\cos\left(\frac{2\pi}{N}nk\right) - j\sum_{n=0}^{N-1} x[n]\sin\left(\frac{2\pi}{N}nk\right)$$

$$X[k] = A[k] - jB[k]$$

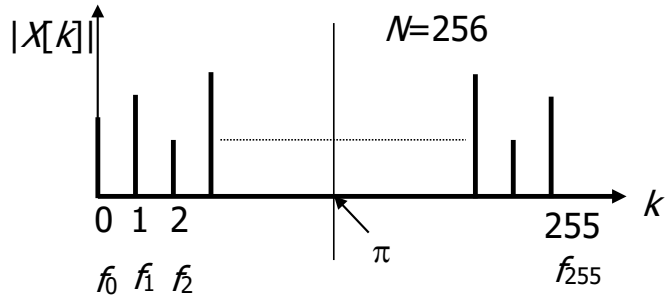
$$|X[k]| = \sqrt{A[k]^2 + B[k]^2} \qquad k = 0,1,2,3,\ldots N-1$$

$$\Delta\theta = \frac{2\pi}{N}$$

$$\Delta f = \frac{f_S}{N}$$

We have $f_k = k\left(\frac{f_s}{N}\right) = k\Delta f \quad (k = 0, 1, 2, \ldots, N-1)$

- N is the length of the sequence x[n] to be analysed using DFT and $|X[k]|$ is the magnitude of the sampled spectrum at index k.



The frequency resolution ($\Delta f = f_s/N$) can be made as small as desired by increasing the value of N.

Example 19

- Let f_s = 8000 Hz, number of sample N = 1000
- DFT bin spacing $\Delta f = \frac{f_S}{N} = \frac{8000}{1000} = 8Hz$
- $f_0 = 0;\ f_1 = 8\text{Hz},\ f_2 = 16\text{ Hz},\ \ldots.\ f_{999} = 7992\text{Hz}$

$$X[k] = \sum_0^{999} x[n]e^{-jn\left(\frac{2\pi k}{1000}\right)},\ k = 0, 1, 2, 3, 4..$$

Example 20

- A speech signal is sampled at a rate of 20000 samples/sec. A sequence of length N = 1024 samples is selected and a 1024-point DFT is computed.

(a) What is the duration of the segment of speech?

Duration = no of samples × sampling period.

= 1024 (1/20000) = 51.2 ms

(b) What is the DFT bin spacing Δf in Hz?

$$\Delta f = \frac{f_S}{N} = \frac{20000}{1024} = 19.531Hz$$

Discrete Fourier Transform (DFT)

Example 21

- Consider the finite length sequence: $x[n] = \delta[n] + 0.2\delta[n-2]$

  Find the *N*-point DFT of *x*[*n*] for *N* = 50.

Answer:

$$X(z) = 1 + 0.2\, z^{-2} \Rightarrow X(\theta) = 1 + 0.2\, e^{-j2\theta}$$

The *N*-point DFT is obtained by evaluating $X(\theta)$ at *N* points equally spaced around the unit circle: $\theta = \frac{2\pi k}{N}$

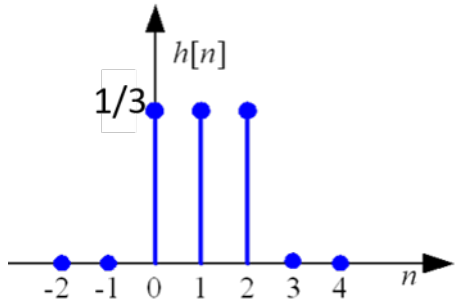
Therefore, the DFT is given by:

$$X[k] = 1 + 0.2e^{-j\left(\frac{2\pi k}{N}\right)2}$$

$$= 1 + 0.2e^{-j\left(\frac{2\pi k}{25}\right)} \qquad k = 0, 1, 2, \ldots. 49$$

Example 22

- Find the 3-point DFT of the three-sample averager. $h[n] = \begin{cases} \frac{1}{3} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$



$$H[k] = \sum_{n=0}^{N-1} h[n]e^{-jn(\frac{2\pi}{N}k)} = \sum_{n=0}^{2} \frac{1}{3} e^{-jn(\frac{2\pi}{N}k)} \quad \textbf{\textit{N=3}}$$

$$H[k] = e^{-j(\frac{2\pi}{N}k)} \frac{(1 + 2\,cos(\frac{2\pi k}{N}))}{3}$$

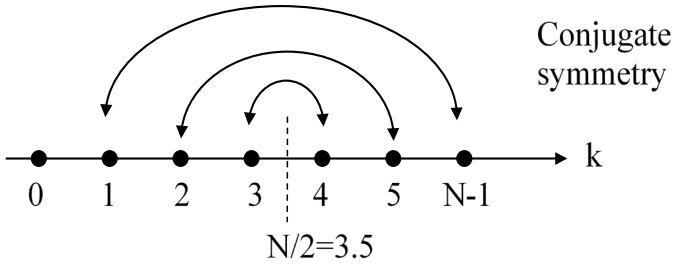
$$|H[k]| = \frac{1}{3}\left|[1 + 2\,cos(\frac{2\pi k}{N})]\right| \quad \text{where } k = 0,1,2$$

At k= 0, the DC gain is |H[0]|= 1

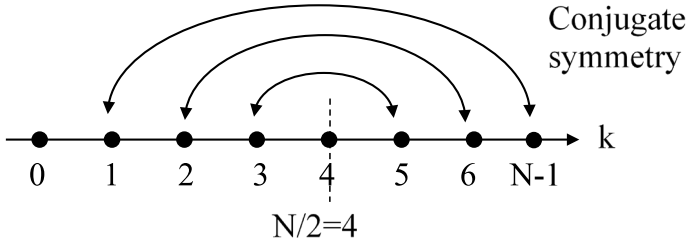
Discrete Fourier Transform (DFT)

Conjugate Symmetry:

- The DFT of a real-valued sequence possesses conjugate symmetry about the origin with $X[-k] = X^*[k]$
- Since the DFT is periodic, $X[-k] = X[N-k]$
- **N is odd** (N=7)



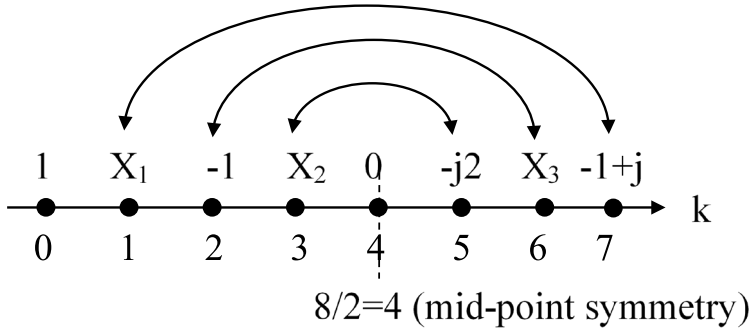
- For odd N, there is no DFT sample at N/2, **X[0] is real.**
- **N is even** (N=8)



- X[0] and X[4] are self-conjugate and therefore **real-valued**.

Example 23

- The DFT of a real signal is $\{1, X_1, -1, X_2, 0, -j2, X_3, -1+j\}$ Find X_1, X_2 and X_3.



$$X_1 = X^*[7] = -1 - j$$

$$X_2 = X^*[5] = 2j$$

$$X_3 = X^*[2] = -1$$

- For N=8, X[0] and X[4] are real.
- **Note:** An 8-point DFT contains 8 frequency samples, and the 8-point IDFT produces 8 time-domain samples.

Discrete Fourier Transform (DFT)

Parseval's Theorem

- Parseval's theorem shows that the signal energy can be computed either in the time domain or from its DFT.
- Therefore, the sum of squares of the signal samples is related to the sum of squares of the magnitude of the DFT samples:

$$\sum_{n=0}^{N-1}|x[n]|^2=\frac{1}{N}\sum_{k=0}^{N-1}|X[k]|^2$$

Example 24

- The DFT of a real signal is: $\{1, X_1, -1, X_2, -7, -j2, X_3, -1+j\}$. What is the signal energy? (Note: First value is at k=0 and the last value is at k = 7)

Ans: X_1, X_2 and X_3 can be found first using conjugate symmetry property

$$\text{Signal Energy}=\frac{1}{8}\sum|X[k]|^2$$

$$=\frac{1}{8}\left[1^2+(\sqrt{2})^2+(-1)^2+2^2+7^2+2^2+1^2+\left(\sqrt{2}\right)^2\right]$$

$$\therefore \text{Signal Energy}=8$$

Inverse Discrete Fourier Transform (IDFT)

- The inverse DFT equation comes directly from the Inverse Fourier Transform equation

$$x[n]=\frac{1}{N}\sum_{k=0}^{N-1}X(k)e^{jn\frac{2\pi}{N}k}\quad 0\le n\le N-1$$

The IDFT recovers the N time-domain samples $x[0], x[1], \ldots, x[N-1]$. If extended beyond this interval, the sequence is periodic with period N

Example 25

- If $X[k] = \{4, -2j, 0, 2j\}$, find its IDFT.

Answer:

N=4, $\therefore x[n]=\frac{1}{4}\sum_{k=0}^{3}X[k]e^{j\frac{2\pi kn}{4}}=\frac{1}{4}\sum_{k=0}^{3}X[k]e^{j\frac{\pi kn}{2}}$

- $n=0 \quad x[0]=\frac{1}{4}[4-j2+0+j2]=1$
- $n=1 \quad x[1]=\frac{1}{4}[4-j2e^{j\frac{\pi}{2}}+0+j2e^{j\frac{3\pi}{2}}]=2$
- $n=2 \quad x[2]=\frac{1}{4}[4-j2e^{j\pi}+0+j2e^{j3\pi}]=1$
- $n=3 \quad x[3]=\frac{1}{4}[4-j2e^{j\frac{3\pi}{2}}+0+j2e^{j\frac{9\pi}{2}}]=0$
- The IDFT is $x[n]=\{1, 2, 1, 0\}$

Discrete Fourier Transform (DFT)

Padding with Zeros and Frequency Resolution

- DFT: $X[k] = \sum_{n=0}^{N-1} x[n] e^{-jn\left(\frac{2\pi k}{N}\right)}$
- The The DFT frequency spacing (Δf) depends on the number of time samples N. For a sampling frequency f_s,

$$\Delta f = \frac{f_s}{N},$$

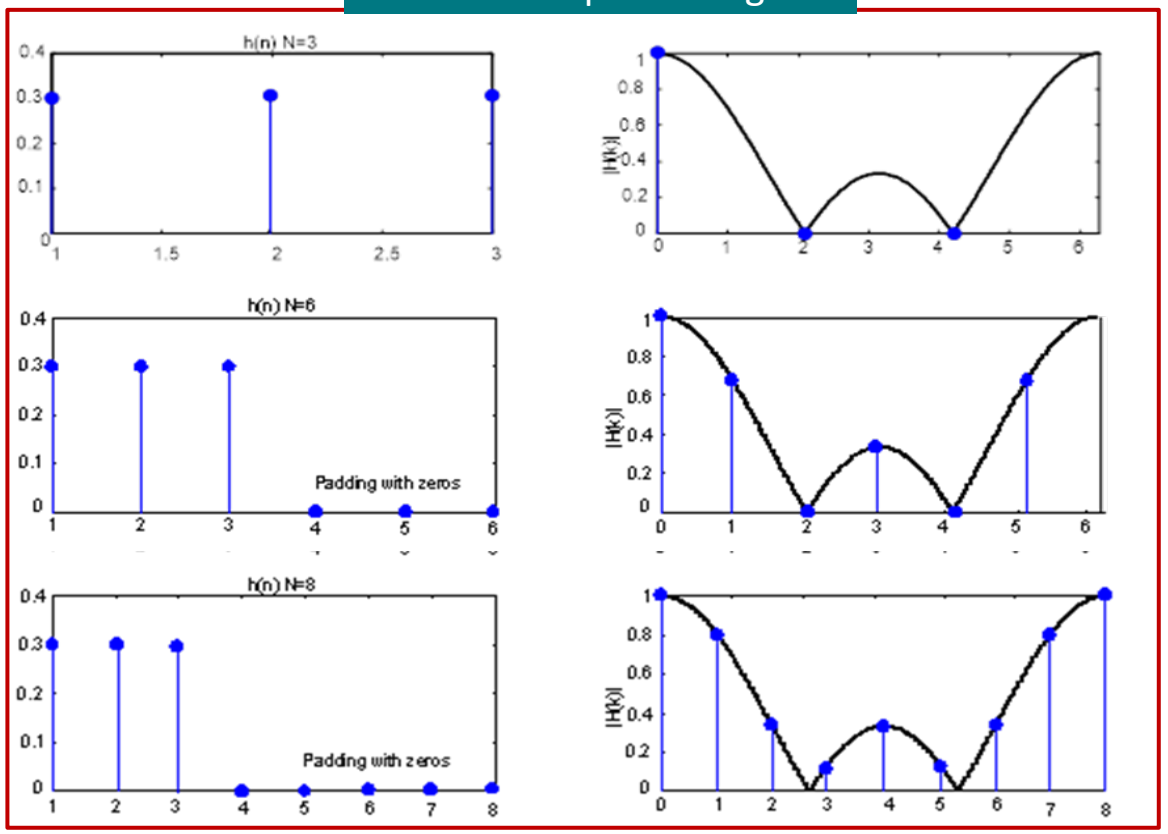
- To obtain more points in the DFT sequence, we can increase the DFT length $N$by appending zero-valued samples to $x[n]$. This procedure is called zero padding.
- The added zeros contribute nothing to the DFT summation, but increasing N decreases the frequency spacing Δf.
- Zero padding gives a denser sampling of the spectrum and provides an interpolated, smoother-looking spectrum for plotting. However, it does not improve the true frequency resolution because no new information is added to the signal.

Example 26

- The three sample averager can be transformed using both the DTFT and the DFT . The DTFT $X(\theta)$ is a continuous function of frequency θ, whereas the DFT consists of discrete samples of $X(\theta)$.
- Adding zeros to the three-sample averager (Example 22) changes the number and spacing of the DFT frequency samples, depending on the DFT length N.
- DFT Lengths of *N* = 3, 6 and 8 are shown on the next slide .
- Although the frequencies at which the DFT is evaluated are different in each case, the underlying spectrum (given by the DTFT) is exactly the same.

Discrete Fourier Transform (DFT)

Three Sample Averager



Zero padding decreases the DFT frequency spacing and provides more samples of the underlying DTFT. Shown here are no zero padding ($N = 3$), padding with 3 zeros ($N = 6$), and padding with 5 zeros ($N = 8$).

Zero padding does not linearly interpolate between the original DFT magnitude samples. It provides additional samples of the underlying continuous DTFT.

Fast Fourier Transform (FFT)

- The Fast Fourier Transform (FFT) is an efficient algorithm for computing the DFT. The commonly used radix-2 FFT requires N to be a power of 2 (Cooley and Tukey, 1965)
- Typically, the DFT may be calculated for a block of **512, 1024, 2048 or more samples**. Direct calculation of the DFT for such large values of Ncan be computationally demanding.
- The FFT exploits **symmetry and redundancy** in the DFT calculation to greatly reduce the computational effort.

  Direct DFT: N^2 operations

  FFT: $N \log_2 N$ *operations*
- The FFT is simply an efficient technique for calculating the DFT - it is not a different transform.

Topic 2 - Summary

Summary

- The definition of a discrete-time system.
- Basic blocks of a discrete-time system: the adder, multiplier and unit delay.
- How to draw the block diagram of a discrete-time system given its difference equation.
- How to write the difference equation of a discrete-time system given its block diagram.
- The definition (with proofs and/or examples) of the following system properties:

  Memoryless (Static), Time Invariant, Linear, Causal and Stable
- Interpretation of a discrete-time signal as a weighted sum of delayed impulses.
- Definition and understanding of convolution (including hand and graphical computation of convolution).
- The impulse response of a linear, time-invariant system, and how to calculate it from a difference equation.
- The difference between an FIR and an IIR system. In particular, the impulse response samples of FIR systems have identical values to the coefficients of the difference equation.

  $y[n] = b0\,x[n] + b1\,x[n-1] + b2\,x[n-2]\ \ ;\ h[n] = \{b0, b1, b2\}.$
- The properties of z-transforms and their applications.
- Convolution in the time domain corresponds to multiplication in the z-domain
- Given the impulse responses of two cascaded LTI systems, compute the overall impulse response using convolution.

Contd…

Topic 2 - Summary

Summary

- How to find the inverse z-transform, given a transfer function.
- Pole-zero description of a discrete -time system.
- Given a pole-zero diagram, transfer function or difference equation, how to find any of the other representations and discuss the system's stability with reference to the pole-zero diagram.
- Determine the frequency response from a transfer function
- Hand-calculate magnitude responses for simple transfer functions and plot them.
- The definition of the Discrete- Time Fourier Transforms (DTFT) and its inverse.
- Given a filter frequency response, find the impulse response using IDTFT.
- The properties of the DTFT and its applications.
- When to use the Discrete Time Fourier transform (DTFT) and when to use the discrete Fourier transform (DFT/FFT).
- Sampling the DTFT (or z-transform on the unit circle) at N equally spaced points to obtain the DFT.
- DFT calculation and conjugate symmetry
- Frequency resolution
- **Zero-padding:** increasing the number of DFT frequency samples without improving the true frequency resolution.